Mathematical Analysis I

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Introduction

Mathematical analysis is one of the fundamental branches of modern mathematics. Its development is closely connected with some of the oldest problems in mathematics: the description of motion, the determination of tangents, the computation of areas and volumes, and the study of continuously varying quantities. These problems gradually led to the creation of differential and integral calculus in the seventeenth century.

The early theory of calculus was extraordinarily successful in computation and application, but its fundamental concepts were not immediately placed on a completely rigorous basis. During the subsequent development of analysis, the notions of number, limit, continuity, derivative, and integral were formulated with increasing precision. In this process, the idea of a limit became one of the central organizing principles of analysis. Many apparently different operations can ultimately be understood through suitable limiting processes.

The purpose of mathematical analysis is therefore not merely to develop techniques for differentiation and integration. Equally important is the logical structure behind these operations. A formula is useful only after the conditions under which it is valid have been understood. For this reason, definitions, theorems, counterexamples, and proofs play a central role throughout the subject.

These notes are organized into three volumes. The first volume begins with the number system and develops the basic theory of limits, functions, derivatives, differentials, and one-variable integration. The second volume passes to multivariable differentiation, multiple integrals, line and surface integrals, applications of integration, and the point-set concepts required for multivariable analysis. The third volume continues with the material corresponding to the later chapters of the course. Together, the three volumes follow the progression from the foundations of one-variable analysis to the subsequent topics of mathematical analysis.

The present volume, *Mathematical Analysis I*, begins with numbers. Rather than taking the real number system merely as an unexplained background, we first examine the basic structure from which the number system is developed. The discussion then proceeds from positive integers and rational numbers to real numbers and their fundamental order properties.

Once the real number system has been established, the notion of limit becomes the main subject. Sequences provide the first setting in which the limiting process can be studied precisely. From convergence and the Cauchy criterion, we proceed to the corresponding theory for functions. Continuity is then introduced as a local property expressed through limits.

Differentiation describes the local change of a function. Starting from the definition of the derivative, we study its basic properties, computational rules, mean value theorems, and the applications of derivatives to the behavior of functions. The differential gives another formulation of local linear change and provides a useful language for subsequent analysis.

Integration is developed from the inverse problem of differentiation. The notion of an antiderivative first leads to indefinite integration and its computational methods. Definite integration is then introduced as a different mathematical construction, and the fundamental theorem of calculus establishes the connection between these two viewpoints. The discussion concludes with the principal properties of integrable functions and the improper integrals contained in the course.

Throughout these notes, definitions and the statements of theorems are kept close to those used in the original course notes and in *Mathematical Analysis* by Gengzhe Chang and

Jihuai Shi. Proofs of the principal results are retained whenever they form an essential part of the theory. Lengthy computational arguments may occasionally be presented as sketches, while the logical assumptions and conclusions of the original results are preserved.

The main purpose of these notes is not to replace the original textbook, but to reorganize the material into a concise record of the course. The emphasis is placed on the definitions, theorems, proofs, and computational principles that form the mathematical structure of the subject, so that previously studied results can be located and reconstructed efficiently.

Chapter 1

Numbers

The number system is the starting point of mathematical analysis. We begin with the positive integers, described by the Peano axioms, and then pass successively to rational numbers and real numbers. The purpose of this chapter is to record the basic properties of numbers that will be used throughout the subsequent theory.

1.1 Peano Axioms and Positive Integers

We denote the set of positive integers by

$$\mathbb{N}^+ = \{1, 2, 3, \ldots\}.$$

The positive integers may be characterized by the Peano axioms.

Definition 1.1.1 (Peano Axioms)**.** The set $\mathbb{N}^+$ is equipped with a distinguished element 1 and a successor operation. The following properties hold:

1. $1 \in \mathbb{N}^+$.
2. Every $n \in \mathbb{N}^+$ has a successor.
3. 1 is not the successor of any positive integer.
4. If two positive integers have the same successor, then they are equal.
5. If a subset $A \subseteq \mathbb{N}^+$ contains 1, and whenever it contains n it also contains the successor of n, then $A = \mathbb{N}^+$.

The last axiom gives the principle of mathematical induction.

Theorem 1.1.2 (Principle of Mathematical Induction)**.** Suppose that a statement $P(n)$ is defined for every $n \in \mathbb{N}^+$. If

1. $P(1)$ is true;
2. whenever $P(n)$ is true, $P(n+1)$ is also true,

then $P(n)$ is true for every $n \in \mathbb{N}^+$.

Proof. Let

$$A = \{n \in \mathbb{N}^+ : P(n) \text{ is true}\}.$$

By the first assumption, $1 \in A$. By the second assumption, if $n \in A$, then $n + 1 \in A$. Hence, by the Peano axioms, $A = \mathbb{N}^+$. Therefore $P(n)$ is true for every $n \in \mathbb{N}^+$. □

The set of integers is denoted by

$$\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}.$$

1.2 Rational Numbers

Definition 1.2.1. A number r is called a rational number if it can be written in the form

$$r = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$ and $q \neq 0$. The set of all rational numbers is denoted by $\mathbb{Q}$.

Rational numbers can be represented by finite decimals or repeating infinite decimals. Conversely, every finite decimal or repeating infinite decimal represents a rational number. For example,

$$0.142857142857\cdots$$

is a repeating decimal. If

$$a = 0.142857142857\cdots,$$

then

$$10^6 a = 142857.142857\cdots,$$

and therefore

$$(10^6 - 1)a = 142857.$$

Hence

$$a = \frac{142857}{999999} = \frac{1}{7}.$$

Not every number is rational. A standard example is $\sqrt{2}$.

Theorem 1.2.2. The number $\sqrt{2}$ is irrational.

Proof. Suppose, to the contrary, that

$$\sqrt{2} = \frac{p}{q},$$

where p and q are positive integers having no common factor. Then

$$p^2 = 2q^2.$$

Thus p^2 is even, and hence p is even. Write $p = 2k$. Then

$$4k^2 = 2q^2,$$

so

$$q^2 = 2k^2.$$

Hence q is also even. Therefore p and q have the common factor 2, contradicting the assumption that the fraction p/q is in lowest terms. Thus $\sqrt{2}$ is irrational. □

The same argument shows that if $n \in \mathbb{N}^+$ is not a perfect square, then $\sqrt{n}$ is irrational.

1.3 Real Numbers

Rational numbers are not sufficient to describe all points on the number line. Irrational numbers such as $\sqrt{2}$ must also be included.

Definition 1.3.1. The collection consisting of all rational numbers and all irrational numbers is called the set of real numbers and is denoted by $\mathbb{R}$.

Thus

$$\mathbb{Q} \subset \mathbb{R}.$$

Every real number can be represented by a decimal expansion. A finite decimal may be regarded as an infinite decimal by appending zeros. Repeating infinite decimals represent rational numbers, while nonrepeating infinite decimals represent irrational numbers.
The correspondence between real numbers and points on a straight line gives the real number line. Choose a point O as the origin, a point P to determine the positive direction and the unit length. Then every real number corresponds to one point on the line, and every point on the line corresponds to one real number.

1.4 Basic Properties of Real Numbers

The real numbers form an ordered field. The basic algebraic and order properties used in these notes are summarized below.
For all $a, b, c \in \mathbb{R}$,

$$a + b = b + a, \qquad ab = ba,$$

and

$$(a + b) + c = a + (b + c), \qquad (ab)c = a(bc).$$

The distributive law is

$$a(b + c) = ab + ac.$$

There exist $0, 1 \in \mathbb{R}$, with $0 \neq 1$, such that

$$a + 0 = a, \qquad a \cdot 1 = a.$$

For every $a \in \mathbb{R}$, there exists $-a \in \mathbb{R}$ such that

$$a + (-a) = 0.$$

For every $a \neq 0$, there exists $a^{-1} \in \mathbb{R}$ such that

$$aa^{-1} = 1.$$

For any $a, b \in \mathbb{R}$, exactly one of

$$a < b, \qquad a = b, \qquad a > b$$

holds.
If $a < b$, then

$$a + c < b + c.$$

If $a < b$ and $c > 0$, then

$$ac < bc.$$

These algebraic and order properties allow the usual operations and inequalities on real numbers.

1.5 Absolute Value

Definition 1.5.1. For $x \in \mathbb{R}$, the absolute value of x is defined by

$$|x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

For all $x, y \in \mathbb{R}$,

$$|x| \geq 0, \qquad |x| = 0 \iff x = 0,$$

and

$$|xy| = |x|\,|y|\,.$$

The triangle inequality is

$$|x + y| \leq |x| + |y|\,.$$

Hence

$$|x - y| \leq |x| + |y|\,.$$

We also have the reverse triangle inequality

$$\big||x| - |y|\big| \leq |x - y|\,.$$

Proof. By the triangle inequality,

$$|x| = |(x - y) + y| \leq |x - y| + |y|\,,$$

so

$$|x| - |y| \leq |x - y|\,.$$

Interchanging x and y gives

$$|y| - |x| \leq |x - y|\,.$$

Therefore

$$\big||x| - |y|\big| \leq |x - y|\,.$$

□

For $a > 0$,

$$|x| < a \quad \Longleftrightarrow \quad -a < x < a,$$

and

$$|x| > a \quad \Longleftrightarrow \quad x > a \quad \text{or} \quad x < -a.$$

1.6 Intervals

Let $a, b \in \mathbb{R}$ with $a < b$.

Definition 1.6.1. The closed interval from a to b is

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Definition 1.6.2. The open interval from a to b is

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

The half-open intervals are

$$[a, b) = \{x \in \mathbb{R} : a \leq x < b\}$$

and

$$(a, b] = \{x \in \mathbb{R} : a < x \leq b\}.$$

Unbounded intervals are written as

$$(-\infty, a), \qquad (-\infty, a], \qquad (a, +\infty), \qquad [a, +\infty).$$

The whole real line is denoted by

$$\mathbb{R} = (-\infty, +\infty).$$

For $x, y \in \mathbb{R}$, the quantity $|x - y|$ is the distance between the corresponding points on the real number line.

1.7 Approximation by Rational Numbers

An important property of the real number system is that rational numbers can approximate real numbers arbitrarily closely.

Theorem 1.7.1. Let $x \in \mathbb{R}$ and let q be a positive integer. Then there exists an integer p such that

$$0 \leq x - \frac{p}{q} < \frac{1}{q}.$$

Proof. For the fixed positive integer q, choose an integer p such that

$$p \leq qx < p + 1.$$

Dividing by $q > 0$, we obtain

$$\frac{p}{q} \leq x < \frac{p+1}{q}.$$

Hence

$$0 \leq x - \frac{p}{q} < \frac{1}{q}.$$

□

Thus, by taking q sufficiently large, a real number can be approximated by rational numbers with arbitrarily small error.

Chapter 2

Limits

The notion of limit is one of the fundamental concepts of mathematical analysis. We begin with sequences, where the limiting process can be described in its simplest form. The theory developed in this chapter will later serve as the basis for limits and continuity of functions.

2.1 Sequences and Convergent Sequences

A sequence is an ordered collection of numbers

$$a_1, a_2, \ldots, a_n, \ldots.$$

It is usually denoted by $\{a_n\}$, where a_n is called the n-th term of the sequence.

Definition 2.1.1. Let $\{a_n\}$ be a sequence. If there exists a real number a such that, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}^+$ such that whenever $n > N$,

$$|a_n - a| < \varepsilon,$$

then the sequence $\{a_n\}$ is said to converge to a, and a is called the limit of $\{a_n\}$.
We write

$$\lim_{n\to\infty} a_n = a$$

or

$$a_n \to a \qquad (n \to \infty).$$

The definition may be written symbolically as

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N}^+, \quad n > N \Longrightarrow |a_n - a| < \varepsilon.$$

The positive integer N depends on ε, but it need not be the smallest positive integer satisfying the condition.

Example 2.1.2. Prove that

$$\lim_{n\to\infty} \frac{3\sqrt{n}+1}{2\sqrt{n}-1} = \frac{3}{2}.$$

Proof. We have

$$\begin{aligned}\left|\frac{3\sqrt{n}+1}{2\sqrt{n}-1} - \frac{3}{2}\right| &= \frac{5}{4\sqrt{n}-2} \\ &< \frac{5}{2\sqrt{n}}.\end{aligned}$$

For any $\varepsilon > 0$, choose N sufficiently large so that

$$\frac{5}{2\sqrt{N}} < \varepsilon.$$

Then $n > N$ implies

$$\left|\frac{3\sqrt{n}+1}{2\sqrt{n}-1} - \frac{3}{2}\right| < \varepsilon.$$

Hence

$$\lim_{n\to\infty} \frac{3\sqrt{n}+1}{2\sqrt{n}-1} = \frac{3}{2}.$$

□

Example 2.1.3. For $a > 1$,

$$\lim_{n\to\infty} \frac{1}{a^n} = 0.$$

Proof. Write $a = 1 + h$, where $h > 0$. By the binomial formula,

$$a^n = (1 + h)^n > 1 + nh > nh.$$

Hence

$$0 < \frac{1}{a^n} < \frac{1}{nh}.$$

Given $\varepsilon > 0$, choose

$$N > \frac{1}{h\varepsilon}.$$

Then $n > N$ implies

$$0 < \frac{1}{a^n} < \frac{1}{nh} < \varepsilon.$$

Therefore

$$\lim_{n\to\infty} \frac{1}{a^n} = 0.$$

□

Example 2.1.4. If $|q| < 1$, then

$$\lim_{n\to\infty} q^n = 0.$$

Example 2.1.5. We have

$$\lim_{n\to\infty} \sqrt[n]{n} = 1.$$

2.2 Properties of Convergent Sequences

Theorem 2.2.1 (Uniqueness of the Limit)**.** If a sequence $\{a_n\}$ converges, then its limit is unique.

Proof. Suppose

$$\lim_{n\to\infty} a_n = a \qquad \text{and} \qquad \lim_{n\to\infty} a_n = b.$$

If $a \neq b$, let

$$\varepsilon = \frac{|a - b|}{3}.$$

For sufficiently large n,

$$|a_n - a| < \varepsilon \qquad \text{and} \qquad |a_n - b| < \varepsilon.$$

Thus

$$|a - b| \leq |a - a_n| + |a_n - b| < 2\varepsilon = \frac{2}{3}\,|a - b|\,,$$

which is impossible. Hence $a = b$. □

Theorem 2.2.2. Every convergent sequence is bounded.

Proof. Suppose

$$a_n \to a.$$

Taking $\varepsilon = 1$, there exists $N \in \mathbb{N}^+$ such that

$$n > N \Longrightarrow |a_n - a| < 1.$$

Hence

$$|a_n| < 1 + |a|$$

for $n > N$.
Let

$$M = \max\{|a_1|, \ldots, |a_N|, 1 + |a|\}.$$

Then

$$|a_n| \leq M$$

for every $n \in \mathbb{N}^+$. Therefore $\{a_n\}$ is bounded. □

Definition 2.2.3. Let

$$a_{k_1}, a_{k_2}, \ldots$$

be terms of the sequence $\{a_n\}$, where

$$k_1 < k_2 < \cdots .$$

Then

$$\{a_{k_n}\}$$

is called a subsequence of $\{a_n\}$.

Theorem 2.2.4. If

$$a_n \to a,$$

then every subsequence of $\{a_n\}$ also converges to a.

Proof. Let $\{a_{k_n}\}$ be a subsequence. Since

$$k_n \geq n,$$

for every $\varepsilon > 0$, once n is sufficiently large, k_n is also sufficiently large. Hence

$$|a_{k_n} - a| < \varepsilon.$$

Therefore

$$a_{k_n} \to a.$$

□

Theorem 2.2.5 (Arithmetic Operations of Limits)**.** Suppose

$$a_n \to a, \qquad b_n \to b.$$

Then

$$\lim_{n\to\infty}(a_n + b_n) = a + b,$$
$$\lim_{n\to\infty}(a_n - b_n) = a - b,$$

and

$$\lim_{n\to\infty} a_n b_n = ab.$$

If $b \neq 0$, then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}.$$

Proof. We prove the first two principal cases.
Since

$$|(a_n + b_n) - (a + b)| \leq |a_n - a| + |b_n - b|,$$

the limit of the sum follows directly from the definition.
For the product,

$$\begin{aligned} |a_n b_n - ab| &= |a_n(b_n - b) + b(a_n - a)| \\ &\leq |a_n|\,|b_n - b| + |b|\,|a_n - a|. \end{aligned}$$

Since a convergent sequence is bounded, $\{a_n\}$ is bounded. The desired conclusion therefore follows from the definition of convergence.
For the quotient, first observe that $b_n \to b \neq 0$ implies that b_n is separated from zero for all sufficiently large n. Thus

$$\frac{1}{b_n} \to \frac{1}{b},$$

and the quotient rule follows from the product rule. □

Definition 2.2.6. A sequence $\{a_n\}$ is called infinitesimal if

$$\lim_{n\to\infty} a_n = 0.$$

Theorem 2.2.7. The sum of finitely many infinitesimal sequences is infinitesimal.

Theorem 2.2.8. The product of an infinitesimal sequence and a bounded sequence is infinitesimal.

Theorem 2.2.9 (Squeeze Theorem)**.** Suppose that

$$a_n \leq b_n \leq c_n$$

for all sufficiently large n, and

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = a.$$

Then

$$\lim_{n\to\infty} b_n = a.$$

Proof. For every $\varepsilon > 0$, when n is sufficiently large,

$$a - \varepsilon < a_n \leq b_n \leq c_n < a + \varepsilon.$$

Hence

$$|b_n - a| < \varepsilon.$$

Therefore $b_n \to a$. □

Theorem 2.2.10. Suppose

$$a_n \to a, \qquad b_n \to b.$$

If $a < b$, then for all sufficiently large n,

$$a_n < b_n.$$

2.3 Generalized Limits of Sequences

The concept of limit can be extended to infinite limits.

Definition 2.3.1. Let $\{a_n\}$ be a sequence. If for every $A > 0$, there exists $N \in \mathbb{N}^+$ such that

$$n > N \Longrightarrow a_n > A,$$

then we write

$$\lim_{n\to\infty} a_n = +\infty.$$

Definition 2.3.2. If for every $A > 0$, there exists $N \in \mathbb{N}^+$ such that

$$n > N \Longrightarrow a_n < -A,$$

then we write

$$\lim_{n\to\infty} a_n = -\infty.$$

If

$$\lim_{n\to\infty} a_n = +\infty$$

or

$$\lim_{n\to\infty} a_n = -\infty,$$

the sequence is called infinitely large.
The notation

$$\lim_{n\to\infty} a_n = \infty$$

means that

$$|a_n| \to +\infty.$$

Remark 2.3.3. A sequence tending to $+\infty$ or $-\infty$ is not a convergent sequence in the ordinary sense.

The usual arithmetic rules for finite and infinite limits may be used whenever the corresponding expression is meaningful. Expressions such as

$$\infty - \infty, \qquad 0 \cdot \infty, \qquad \frac{\infty}{\infty}$$

cannot be determined directly from the symbols alone.

2.4 Monotone Sequences

Definition 2.4.1. A sequence $\{a_n\}$ is called increasing if

$$a_n \leq a_{n+1} \qquad (n = 1, 2, \ldots).$$

It is called strictly increasing if

$$a_n < a_{n+1}.$$

Definition 2.4.2. A sequence $\{a_n\}$ is called decreasing if

$$a_n \geq a_{n+1} \qquad (n = 1, 2, \ldots).$$

It is called strictly decreasing if

$$a_n > a_{n+1}.$$

Increasing and decreasing sequences are collectively called monotone sequences.

Theorem 2.4.3 (Monotone Convergence Theorem)**.** A monotone sequence has a finite limit if and only if it is bounded.

Proof. Necessity follows from the fact that every convergent sequence is bounded. Suppose first that $\{a_n\}$ is increasing and bounded above. Let

$$a = \sup\{a_n : n \in \mathbb{N}^+\}.$$

For every $\varepsilon > 0$, $a - \varepsilon$ cannot be an upper bound. Hence there exists $N \in \mathbb{N}^+$ such that

$$a_N > a - \varepsilon.$$

Since the sequence is increasing,

$$a - \varepsilon < a_N \leq a_n \leq a$$

for every $n > N$. Therefore

$$|a_n - a| < \varepsilon,$$

and hence

$$a_n \to a.$$

The decreasing case is proved similarly by taking the infimum. □

Example 2.4.4. The sequence

$$a_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$$

is increasing but unbounded.

Proof. For $n = 2^k$,

$$\begin{aligned} a_{2^k} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \cdots \\ + \left(\frac{1}{2^{k-1}+1} + \cdots + \frac{1}{2^k}\right). \end{aligned}$$

Each parenthesis after the first contains 2^{j-1} terms, each not smaller than $1/2^j$. Thus every such parenthesis is at least $1/2$. Hence

$$a_{2^k} \geq 1 + \frac{k}{2}.$$

Therefore the sequence is unbounded. □

Example 2.4.5. For $n \in \mathbb{N}^+$, let

$$a_n = 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^n}.$$

Then $\{a_n\}$ is increasing and bounded above, and hence convergent.

Theorem 2.4.6 (Nested Interval Theorem)**.** Let

$$I_n = [a_n, b_n], \qquad n \in \mathbb{N}^+,$$

be a sequence of closed intervals such that

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

and

$$b_n - a_n \to 0.$$

Then there exists a unique real number a such that

$$a \in I_n$$

for every $n \in \mathbb{N}^+$.

Proof. Since the intervals are nested,

$$a_1 \leq a_2 \leq \cdots \leq b_2 \leq b_1.$$

Thus $\{a_n\}$ is increasing and bounded above, while $\{b_n\}$ is decreasing and bounded below. Hence both converge. Let

$$\lim_{n\to\infty} a_n = a, \qquad \lim_{n\to\infty} b_n = b.$$

Since

$$b_n - a_n \to 0,$$

we have $a = b$. Moreover,

$$a_n \leq a \leq b_n$$

for every n, so $a \in I_n$.
If another number c belongs to every I_n, then

$$|a - c| \leq b_n - a_n$$

for every n. Letting $n \to \infty$ gives $a = c$. □

2.5 The Number e

Consider the sequence

$$e_n = \left(1 + \frac{1}{n}\right)^n.$$

Using the binomial formula,

$$\begin{aligned} e_n = 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) \\ + \frac{1}{3!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right) + \cdots. \end{aligned}$$

The sequence $\{e_n\}$ is increasing. Moreover,

$$e_n < 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!} < 3.$$

Hence, by the monotone convergence theorem, $\{e_n\}$ has a finite limit.

Definition 2.5.1. The number e is defined by

$$e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

Numerically,

$$e = 2.7182818\ldots.$$

The sequence

$$s_n = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!}$$

also converges to e. Thus

$$e = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \cdots.$$

Theorem 2.5.2. The number e is irrational.

Proof. Suppose, to the contrary, that

$$e = \frac{p}{q}, \qquad p, q \in \mathbb{N}^+.$$

Take $n = q$. Since

$$e = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{q!} + \frac{1}{(q+1)!} + \cdots,$$

multiplication by $q!$ gives

$$q!e = q!\left(1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{q!}\right) + q!\sum_{k=q+1}^{\infty} \frac{1}{k!}.$$

The first two terms on the two sides are integers, so

$$q!\sum_{k=q+1}^{\infty} \frac{1}{k!}$$

must be an integer.
However,

$$0 < q!\sum_{k=q+1}^{\infty} \frac{1}{k!} < \frac{1}{q}\left(1 + \frac{1}{q} + \frac{1}{q^2} + \cdots\right) < 1,$$

which is impossible. Hence e is irrational. □

Several standard limits follow from the definition of e, including

$$\lim_{n\to\infty} \left(1 - \frac{1}{n}\right)^n = \frac{1}{e}$$

and

$$\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = e^a$$

in the cases considered in the course.

2.6 Fundamental Sequences and the Cauchy Criterion

Definition 2.6.1. A sequence $\{a_n\}$ is called a fundamental sequence, or a Cauchy sequence, if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}^+$ such that

$$m, n > N \Longrightarrow |a_n - a_m| < \varepsilon.$$

Equivalently, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}^+$ such that

$$n > N, \quad p \in \mathbb{N}^+ \Longrightarrow |a_{n+p} - a_n| < \varepsilon.$$

Theorem 2.6.2. Every convergent sequence is a fundamental sequence.

Proof. Suppose

$$a_n \to a.$$

For every $\varepsilon > 0$, choose N such that

$$n > N \Longrightarrow |a_n - a| < \frac{\varepsilon}{2}.$$

If $m, n > N$, then

$$\begin{aligned} |a_n - a_m| &\le |a_n - a| + |a_m - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Thus $\{a_n\}$ is a fundamental sequence. □

Theorem 2.6.3 (Bolzano–Weierstrass Theorem)**.** Every bounded sequence has a convergent subsequence.

Sketch of Proof. Let $\{a_n\}$ be bounded. Then all its terms lie in some closed interval I_1. Divide I_1 into two equal closed intervals. At least one of them contains infinitely many terms of the sequence; denote such an interval by I_2.
Repeat this process. We obtain nested closed intervals

$$I_1 \supseteq I_2 \supseteq \cdots,$$

each containing infinitely many terms of the sequence, and their lengths tend to zero.
By the nested interval theorem, there is a unique point a common to all these intervals. Choose successively

$$a_{n_k} \in I_k, \qquad n_1 < n_2 < \cdots.$$

Since the length of I_k tends to zero,

$$a_{n_k} \to a.$$

□

Theorem 2.6.4 (Cauchy Convergence Criterion)**.** A sequence of real numbers converges if and only if it is a fundamental sequence.

Proof. The necessity has already been proved.
Suppose now that $\{a_n\}$ is a fundamental sequence. Taking $\varepsilon = 1$, there exists N such that

$$m, n > N \Longrightarrow |a_n - a_m| < 1.$$

Fixing $m = N + 1$, we obtain

$$|a_n| \leq 1 + |a_{N+1}|$$

for all sufficiently large n. Hence $\{a_n\}$ is bounded.
By the Bolzano–Weierstrass theorem, $\{a_n\}$ has a convergent subsequence

$$a_{n_k} \to a.$$

For every $\varepsilon > 0$, choose N_1 such that

$$m, n > N_1 \Longrightarrow |a_n - a_m| < \frac{\varepsilon}{2}.$$

Choose k sufficiently large so that

$$n_k > N_1 \qquad \text{and} \qquad |a_{n_k} - a| < \frac{\varepsilon}{2}.$$

Then for $n > N_1$,

$$\begin{aligned} |a_n - a| &\leq |a_n - a_{n_k}| + |a_{n_k} - a| \\ &< \varepsilon. \end{aligned}$$

Therefore

$$a_n \to a.$$

□

2.7 Supremum and Infimum

Let $E \subset \mathbb{R}$ be nonempty.

Definition 2.7.1. A number $M \in \mathbb{R}$ is called an upper bound of E if

$$x \leq M$$

for every $x \in E$.

Definition 2.7.2. A number $m \in \mathbb{R}$ is called a lower bound of E if

$$m \leq x$$

for every $x \in E$.

If E has an upper bound, then E is called bounded above. If E has a lower bound, then E is called bounded below.

Definition 2.7.3. Let E be nonempty and bounded above. A number α is called the supremum of E if

1. $x \leq \alpha$ for every $x \in E$;
2. for every $\varepsilon > 0$, there exists $x \in E$ such that

$$x > \alpha - \varepsilon.$$

We write

$$\alpha = \sup E.$$

Definition 2.7.4. Let E be nonempty and bounded below. A number β is called the infimum of E if

1. $\beta \leq x$ for every $x \in E$;
2. for every $\varepsilon > 0$, there exists $x \in E$ such that

$$x < \beta + \varepsilon.$$

We write

$$\beta = \inf E.$$

Theorem 2.7.5 (Supremum Principle)**.** Every nonempty set of real numbers that is bounded above has a supremum.

Sketch of Proof. Let E be nonempty and bounded above. Choose an interval $[a_1, b_1]$ with $a_1 \in E$ and b_1 an upper bound of E.
Bisect the interval. If the midpoint is an upper bound, retain the left half; otherwise retain the right half. Repeating the procedure, we obtain nested intervals

$$[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$$

whose lengths tend to zero, where each a_n is not an upper bound and each b_n is an upper bound.
By the nested interval theorem, there is a unique point α common to all intervals. Since every b_n is an upper bound, α is an upper bound. Since every a_n is not an upper bound and

$$b_n - a_n \to 0,$$

points of E can be found arbitrarily close to α from below. Hence

$$\alpha = \sup E.$$

□

Corollary 2.7.6 (Infimum Principle)**.** Every nonempty set of real numbers that is bounded below has an infimum.

For convenience, if E is not bounded above, we write

$$\sup E = +\infty.$$

If E is not bounded below, we write

$$\inf E = -\infty.$$

2.8 Finite Covering Theorem

Definition 2.8.1. Let $A \subset \mathbb{R}$. A family of open intervals

$$\{I_\lambda\}_{\lambda \in \Lambda}$$

is called an open covering of A if

$$A \subseteq \bigcup_{\lambda \in \Lambda} I_\lambda.$$

If finitely many intervals from the covering already cover A, they are called a finite subcovering.

Theorem 2.8.2 (Finite Covering Theorem)**.** Let $[a, b]$ be a closed interval. Every open covering of $[a, b]$ contains a finite subcovering.

Proof. Suppose, to the contrary, that an open covering of $[a, b]$ has no finite subcovering. Bisect $[a, b]$. At least one of the two closed subintervals cannot be covered by finitely many intervals from the given covering. Denote such an interval by I_1.
Bisect I_1 and repeat the argument. We obtain nested closed intervals

$$I_1 \supseteq I_2 \supseteq \cdots$$

such that no I_n has a finite subcovering and

$$|I_n| \to 0.$$

By the nested interval theorem, there exists a unique point

$$\xi \in \bigcap_{n=1}^{\infty} I_n.$$

Since the original family covers $[a, b]$, there is an open interval I of the covering such that $\xi \in I$.
Because I is open, there exists $\delta > 0$ such that

$$(\xi - \delta, \xi + \delta) \subset I.$$

For sufficiently large n, the interval I_n has length less than δ and contains ξ, so

$$I_n \subset I.$$

Thus I_n is covered by one member of the covering, contradicting its construction. Therefore a finite subcovering must exist. □

2.9 Upper and Lower Limits

Let $\{a_n\}$ be a sequence. For each n, consider the tail set

$$E_n = \{a_n, a_{n+1}, a_{n+2}, \ldots\}.$$

Define

$$\alpha_n = \sup E_n, \qquad \beta_n = \inf E_n.$$

Then

$$\alpha_1 \geq \alpha_2 \geq \cdots, \qquad \beta_1 \leq \beta_2 \leq \cdots.$$

Definition 2.9.1. The upper limit and lower limit of $\{a_n\}$ are defined by

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} \alpha_n = \lim_{n\to\infty} \sup_{k\geq n} a_k$$

and

$$\liminf_{n\to\infty} a_n = \lim_{n\to\infty} \beta_n = \lim_{n\to\infty} \inf_{k\geq n} a_k.$$

The upper and lower limits may take the values

$$+\infty \qquad \text{or} \qquad -\infty.$$

Theorem 2.9.2. Let

$$a^* = \limsup_{n\to\infty} a_n.$$

Then:

1. there exists a subsequence of $\{a_n\}$ converging to a^*;
2. no subsequential limit of $\{a_n\}$ is greater than a^*.

Thus a^* is the greatest subsequential limit.

Similarly,

$$a_* = \liminf_{n\to\infty} a_n$$

is the least subsequential limit.

Theorem 2.9.3. A sequence $\{a_n\}$ converges to a if and only if

$$\limsup_{n\to\infty} a_n = \liminf_{n\to\infty} a_n = a.$$

Theorem 2.9.4. For sequences $\{a_n\}$ and $\{b_n\}$,

$$\liminf_{n\to\infty} a_n + \liminf_{n\to\infty} b_n \le \liminf_{n\to\infty}(a_n + b_n),$$

and

$$\limsup_{n\to\infty}(a_n + b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n,$$

whenever the expressions involved are meaningful.

2.10 Stolz Theorem

The Stolz theorem is particularly useful for limits having a discrete form analogous to the indeterminate forms

$$\frac{\infty}{\infty} \qquad \text{and} \qquad \frac{0}{0}.$$

Theorem 2.10.1 (Stolz Theorem, ∞/∞ Form)**.** Let $\{b_n\}$ be strictly increasing and

$$b_n \to +\infty.$$

If

$$\lim_{n\to\infty} \frac{a_n - a_{n-1}}{b_n - b_{n-1}} = A,$$

where A may be finite or infinite, then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Proof. We first suppose A is finite. Given $\varepsilon > 0$, for sufficiently large n,

$$A - \varepsilon < \frac{a_n - a_{n-1}}{b_n - b_{n-1}} < A + \varepsilon.$$

Since $b_n - b_{n-1} > 0$,

$$(A-\varepsilon)(b_n - b_{n-1}) < a_n - a_{n-1} < (A+\varepsilon)(b_n - b_{n-1}).$$

Fix n_0 sufficiently large and sum from n_0+1 to n. Then

$$\begin{aligned}(A-\varepsilon)(b_n - b_{n_0}) &< a_n - a_{n_0} \\ &< (A+\varepsilon)(b_n - b_{n_0}).\end{aligned}$$

Dividing by $b_n > 0$,

$$\begin{aligned}(A-\varepsilon)\left(1 - \frac{b_{n_0}}{b_n}\right) + \frac{a_{n_0}}{b_n} &< \frac{a_n}{b_n} \\ &< (A+\varepsilon)\left(1 - \frac{b_{n_0}}{b_n}\right) + \frac{a_{n_0}}{b_n}.\end{aligned}$$

Since $b_n \to +\infty$,

$$\frac{b_{n_0}}{b_n} \to 0, \qquad \frac{a_{n_0}}{b_n} \to 0.$$

Therefore

$$A - \varepsilon \le \liminf_{n\to\infty} \frac{a_n}{b_n} \le \limsup_{n\to\infty} \frac{a_n}{b_n} \le A + \varepsilon.$$

Letting $\varepsilon \to 0$, we obtain

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

The cases $A = +\infty$ and $A = -\infty$ are proved in the same way. □

Corollary 2.10.2. If

$$x_n \to a,$$

then

$$\lim_{n\to\infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = a.$$

Proof. Let

$$a_n = x_1 + x_2 + \cdots + x_n, \qquad b_n = n.$$

Then

$$\frac{a_n - a_{n-1}}{b_n - b_{n-1}} = x_n \to a.$$

The result follows from the Stolz theorem. □

Example 2.10.3. For a positive integer k,

$$\lim_{n\to\infty} \frac{1^k + 2^k + \cdots + n^k}{n^{k+1}} = \frac{1}{k+1}.$$

Proof. Set

$$a_n = 1^k + 2^k + \cdots + n^k, \qquad b_n = n^{k+1}.$$

Then

$$a_n - a_{n-1} = n^k$$

and

$$b_n - b_{n-1} = n^{k+1} - (n-1)^{k+1}.$$

Therefore

$$\frac{a_n - a_{n-1}}{b_n - b_{n-1}} = \frac{n^k}{n^{k+1} - (n-1)^{k+1}} \to \frac{1}{k+1}.$$

By the Stolz theorem,

$$\lim_{n\to\infty} \frac{1^k + 2^k + \cdots + n^k}{n^{k+1}} = \frac{1}{k+1}.$$

□

Theorem 2.10.4 (Stolz Theorem, 0/0 Form)**.** Suppose

$$a_n \to 0, \qquad b_n \to 0,$$

and

$$b_n > b_{n+1} > 0.$$

If

$$\lim_{n\to\infty} \frac{a_n - a_{n+1}}{b_n - b_{n+1}} = A,$$

then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = A.$$

Chapter 3

Cardinality of Sets

The concept of cardinality is used to compare the sizes of sets. For finite sets, the meaning of "having the same number of elements" is immediate. For infinite sets, however, counting the elements directly is no longer possible. The comparison is therefore made through one-to-one correspondences between sets.

3.1 Sets of the Same Cardinality

Definition 3.1.1. Let A and B be two sets. If there exists a one-to-one mapping from A onto B, then A and B are said to have the same cardinality.

In other words, A and B have the same cardinality if their elements can be put into a one-to-one correspondence.
This relation has the following properties:

1. A has the same cardinality as A;
2. if A has the same cardinality as B, then B has the same cardinality as A;
3. if A has the same cardinality as B, and B has the same cardinality as C, then A has the same cardinality as C.

The first property is reflexivity, the second is symmetry, and the third is transitivity.

3.2 Finite and Infinite Sets

Let

$$N_n = \{1, 2, \ldots, n\}.$$

Definition 3.2.1. If there exists a positive integer n such that a set A has the same cardinality as N_n, then A is called a finite set.

The empty set is also regarded as a finite set.

Definition 3.2.2. A set that is not finite is called an infinite set.

Definition 3.2.3. If an infinite set has the same cardinality as $\mathbb{N}^+$, then it is called a countable set.

Thus a countable set can be arranged in the form

$$a_1, a_2, a_3, \ldots.$$

Definition 3.2.4. An infinite set that is not countable is called an uncountable set.

Hence every set belongs to one of the following three classes:

1. finite sets;
2. countable sets;
3. uncountable sets.

3.3 Infinite Sets and Countable Subsets

Theorem 3.3.1. Every infinite set contains a countable subset.

Proof. Let A be an infinite set. Choose an element $a_1 \in A$. Since A is infinite,

$$A \setminus \{a_1\}$$

is nonempty, so we may choose

$$a_2 \in A \setminus \{a_1\}.$$

Suppose that distinct elements

$$a_1, a_2, \ldots, a_{k-1}$$

have already been chosen. Since A is infinite,

$$A \setminus \{a_1, a_2, \ldots, a_{k-1}\}$$

is nonempty. Choose

$$a_k \in A \setminus \{a_1, a_2, \ldots, a_{k-1}\}.$$

Continuing in this way, we obtain a sequence of distinct elements

$$a_1, a_2, a_3, \ldots.$$

Therefore

$$\{a_1, a_2, a_3, \ldots\}$$

is a countable subset of A. □

This theorem shows that countable sets are, in a sense, the "smallest" infinite sets.

3.4 Countable Unions of Countable Sets

Theorem 3.4.1. Let

$$S = E_1 \cup E_2 \cup E_3 \cup \cdots,$$

where each E_n is a countable set. Then S is countable.

Proof. For each $n \in \mathbb{N}^+$, write

$$E_n = \{x_{n1}, x_{n2}, x_{n3}, \ldots\}.$$

Arrange the elements in an infinite array:

$$\begin{array}{cccccc} x_{11} & x_{12} & x_{13} & x_{14} & \cdots \\ x_{21} & x_{22} & x_{23} & x_{24} & \cdots \\ x_{31} & x_{32} & x_{33} & x_{34} & \cdots \\ x_{41} & x_{42} & x_{43} & x_{44} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \end{array}$$

and list the elements along successive diagonals:

$$x_{11}, x_{12}, x_{21}, x_{31}, x_{22}, x_{13}, \ldots.$$

If the same element appears more than once, retain only its first appearance. In this way all elements of S can be arranged in a sequence. Hence S is countable. □

3.5 The Rational Numbers Are Countable

Theorem 3.5.1. The set of positive rational numbers is countable.

Proof. Every positive rational number can be written as

$$\frac{p}{q}, \qquad p, q \in \mathbb{N}^+.$$

Arrange these fractions according to the value of $p + q$:

$$\frac{1}{1}; \qquad \frac{1}{2}, \frac{2}{1}; \qquad \frac{1}{3}, \frac{2}{2}, \frac{3}{1}; \qquad \dots.$$

Starting from the first row, list the fractions in order and delete repeated values. Every positive rational number eventually occurs in this list. Therefore the set of positive rational numbers is countable. □

Corollary 3.5.2. The set $\mathbb{Q}$ of all rational numbers is countable.

Proof. We have

$$\mathbb{Q} = \mathbb{Q}_+ \cup \{0\} \cup (-\mathbb{Q}_+).$$

The positive rational numbers and the negative rational numbers are both countable, while $\{0\}$ is finite. Hence $\mathbb{Q}$ is countable. □

3.6 The Interval $(0, 1)$ Is Uncountable

Theorem 3.6.1. The set of real numbers in the interval $(0, 1)$ is uncountable.

Proof. Suppose, to the contrary, that $(0, 1)$ is countable. Then its elements can be arranged as

$$x_1, x_2, x_3, \dots.$$

Divide $[0, 1]$ into three equal closed intervals. At least one of these intervals does not contain x_1. Choose such an interval and denote it by I_1.
Next divide I_1 into three equal closed intervals. At least one of them does not contain x_2. Choose such an interval and denote it by I_2.
Continue this construction. We obtain nested closed intervals

$$I_1 \supset I_2 \supset I_3 \supset \cdots$$

such that

$$x_n \notin I_n$$

for every n, and the length of I_n is

$$\frac{1}{3^n}.$$

By the nested interval theorem, there exists a point x^* belonging to every I_n:

$$x^* \in \bigcap_{n=1}^{\infty} I_n.$$

Since $x_n \notin I_n$, we have

$$x^* \neq x_n$$

for every $n \in \mathbb{N}^+$.
Thus $x^* \in (0, 1)$, but x^* does not occur in the assumed list

$$x_1, x_2, x_3, \dots.$$

This is a contradiction. Therefore $(0, 1)$ is uncountable. □

Corollary 3.6.2. The closed interval $[0, 1]$ is uncountable.

Indeed, $(0, 1) \subset [0, 1]$, so if $[0, 1]$ were countable, then $(0, 1)$ would also be countable.
The textbook also gives a direct one-to-one correspondence between $[0, 1]$ and $(0, 1)$. Hence these two intervals have the same cardinality.

3.7 Countable and Uncountable Sets

The preceding results give the basic examples needed in these notes:

$$\mathbb{N}^+ \quad \text{is countable,}$$

$$\mathbb{Q} \quad \text{is countable,}$$

whereas

$$(0, 1) \quad \text{and} \quad [0, 1]$$

are uncountable.
Thus infinite sets need not have the same cardinality. In particular, the set of real numbers in an interval cannot be listed in a sequence in the way that the rational numbers can.

Chapter 4

Mappings and Functions

The concept of a mapping describes a correspondence between two sets. Functions are mappings whose domain and range consist of numbers. In this chapter, we begin with mappings and functions, then develop limits and continuity of functions. The later sections concern uniform continuity, properties of continuous functions on closed intervals, upper and lower limits, and the iteration of functions.

4.1 Mappings of Sets

Definition 4.1.1. Let A and B be two sets. If there is a rule f such that for every $x \in A$, there is a uniquely determined element $f(x) \in B$, then f is called a mapping from A to B, written as

$$f : A \to B.$$

The set A is called the domain of f, and $f(x)$ is called the image of x under f.

If $E \subseteq A$, the image of E under f is defined by

$$f(E) = \{f(x) : x \in E\}.$$

If $F \subseteq B$, the inverse image of F under f is defined by

$$f^{-1}(F) = \{x \in A : f(x) \in F\}.$$

Definition 4.1.2. A mapping $f : A \to B$ is called an onto mapping if

$$f(A) = B.$$

Definition 4.1.3. A mapping $f : A \to B$ is called one-to-one if

$$f(x_1) = f(x_2)$$

implies

$$x_1 = x_2.$$

A mapping that is both one-to-one and onto is called a one-to-one correspondence.

Definition 4.1.4. Let

$$f : A \to B, \qquad g : B \to C.$$

The composite mapping $g \circ f : A \to C$ is defined by

$$(g \circ f)(x) = g(f(x)).$$

In general,

$$f \circ g \neq g \circ f.$$

Composition of mappings satisfies the associative law:

$$(f \circ g) \circ h = f \circ (g \circ h),$$

whenever both sides are defined.

Definition 4.1.5. Suppose $f : A \to B$ is one-to-one and onto. Then its inverse mapping

$$f^{-1} : B \to A$$

is defined by

$$f^{-1}(y) = x \quad \Longleftrightarrow \quad f(x) = y.$$

Thus

$$f^{-1} \circ f = I_A$$

and

$$f \circ f^{-1} = I_B,$$

where I_A and I_B are the identity mappings on A and B, respectively.

4.2 Functions

A function is a special type of mapping.

Definition 4.2.1. Let X and Y be sets of real numbers. A mapping

$$f : X \to Y$$

is called a real-valued function of a real variable.

The set X is the domain of the function. If $x \in X$, then $f(x)$ is the value of f at x.
In many problems a formula is used to define a function. In this case, the domain is the collection of all values of x for which the expression is meaningful, unless the domain is specified separately.
For two functions f and g, their sum, difference, product, and quotient are defined by

$$(f+g)(x) = f(x) + g(x),$$

$$(f-g)(x) = f(x) - g(x),$$

$$(fg)(x) = f(x)g(x),$$

and

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)},$$

where the expressions are defined.

Definition 4.2.2. A function f is called increasing on an interval I if for any $x_1, x_2 \in I$,

$$x_1 < x_2 \Longrightarrow f(x_1) \leq f(x_2).$$

If

$$x_1 < x_2 \Longrightarrow f(x_1) < f(x_2),$$

then f is called strictly increasing.

Decreasing and strictly decreasing functions are defined similarly.

Theorem 4.2.3. Suppose $f : X \to Y$ is strictly increasing or strictly decreasing. Then f is one-to-one and hence has an inverse function on its range.

If f is strictly increasing, then f^{-1} is also strictly increasing. If f is strictly decreasing, then f^{-1} is also strictly decreasing.

4.3 Limits of Functions

Let f be defined in a deleted neighborhood of x_0.

Definition 4.3.1. A real number l is called the limit of $f(x)$ as $x \to x_0$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - x_0| < \delta$$

implies

$$|f(x) - l| < \varepsilon.$$

In this case, we write

$$\lim_{x \to x_0} f(x) = l.$$

The value $f(x_0)$ does not enter into the definition of the limit. Indeed, f need not even be defined at x_0.

Example 4.3.2. We have

$$\lim_{x\to 0} \sin x = 0.$$

Proof. Since

$$|\sin x| \le |x|,$$

given $\varepsilon > 0$, choose $\delta = \varepsilon$. Then

$$0 < |x| < \delta$$

implies

$$|\sin x| < \varepsilon.$$

Therefore

$$\lim_{x\to 0} \sin x = 0.$$

□

Theorem 4.3.3 (Heine Theorem)**.** The equality

$$\lim_{x\to x_0} f(x) = l$$

holds if and only if for every sequence $\{x_n\}$ satisfying

$$x_n \neq x_0, \qquad x_n \to x_0,$$

we have

$$f(x_n) \to l.$$

Proof. Suppose first that

$$\lim_{x\to x_0} f(x) = l.$$

Given $\varepsilon > 0$, choose $\delta > 0$ such that

$$0 < |x - x_0| < \delta \Longrightarrow |f(x) - l| < \varepsilon.$$

Since $x_n \to x_0$, for sufficiently large n,

$$0 < |x_n - x_0| < \delta.$$

Hence

$$|f(x_n) - l| < \varepsilon,$$

so

$$f(x_n) \to l.$$

Conversely, suppose the limit does not equal l. Then there exists $\varepsilon_0 > 0$ such that for every positive integer n, one can choose x_n satisfying

$$0 < |x_n - x_0| < \frac{1}{n}$$

and

$$|f(x_n) - l| \ge \varepsilon_0.$$

Then $x_n \to x_0$, but $f(x_n) \not\to l$, a contradiction. □

Theorem 4.3.4 (Uniqueness of Functional Limits)**.** If

$$\lim_{x \to x_0} f(x)$$

exists, then it is unique.

Theorem 4.3.5. If

$$\lim_{x \to x_0} f(x) = l,$$

then f is bounded in some deleted neighborhood of x_0.

Theorem 4.3.6 (Arithmetic Operations of Limits)**.** Suppose

$$\lim_{x \to x_0} f(x) = l, \qquad \lim_{x \to x_0} g(x) = m.$$

Then

$$\lim_{x \to x_0} (f(x) + g(x)) = l + m,$$

$$\lim_{x \to x_0} f(x)g(x) = lm,$$

and, if $m \neq 0$,

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{l}{m}.$$

Theorem 4.3.7 (Squeeze Theorem)**.** Suppose

$$f(x) \leq h(x) \leq g(x)$$

in a deleted neighborhood of x_0, and

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} g(x) = l.$$

Then

$$\lim_{x \to x_0} h(x) = l.$$

Theorem 4.3.8 (Cauchy Criterion)**.** The finite limit

$$\lim_{x \to x_0} f(x)$$

exists if and only if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever

$$0 < |x_1 - x_0| < \delta, \qquad 0 < |x_2 - x_0| < \delta,$$

we have

$$|f(x_1) - f(x_2)| < \varepsilon.$$

4.4 One-Sided Limits

Definition 4.4.1. If for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < x - x_0 < \delta$$

implies

$$|f(x) - l| < \varepsilon,$$

then l is called the right-hand limit of f at x_0, written as

$$\lim_{x \to x_0+} f(x) = l.$$

Similarly, if

$$0 < x_0 - x < \delta$$

implies

$$|f(x) - l| < \varepsilon,$$

then

$$\lim_{x \to x_0-} f(x) = l$$

is called the left-hand limit.

Theorem 4.4.2. The finite limit

$$\lim_{x \to x_0} f(x) = l$$

exists if and only if both one-sided limits exist and

$$\lim_{x \to x_0-} f(x) = \lim_{x \to x_0+} f(x) = l.$$

A corresponding statement holds for infinite limits.

Example 4.4.3. The fundamental trigonometric limit is

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

Proof. For $0 < x < \pi/2$, the geometric argument gives

$$\sin x < x < \tan x.$$

Dividing by $\sin x > 0$,

$$1 < \frac{x}{\sin x} < \frac{1}{\cos x}.$$

Hence

$$\cos x < \frac{\sin x}{x} < 1.$$

Since

$$\lim_{x \to 0} \cos x = 1,$$

the squeeze theorem gives

$$\lim_{x \to 0+} \frac{\sin x}{x} = 1.$$

Since $\sin x/x$ is even, the left-hand limit has the same value. Therefore

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

□

4.5 Other Forms of Limit Processes

The definition of a limit can be adapted to the cases $x \to +\infty$, $x \to -\infty$, and infinite limits.

Definition 4.5.1. We write

$$\lim_{x \to +\infty} f(x) = l$$

if for every $\varepsilon > 0$, there exists $A > 0$ such that

$$x > A \Longrightarrow |f(x) - l| < \varepsilon.$$

Similarly,

$$\lim_{x \to -\infty} f(x) = l$$

means that for every $\varepsilon > 0$, there exists $A > 0$ such that

$$x < -A \Longrightarrow |f(x) - l| < \varepsilon.$$

Theorem 4.5.2. The equality

$$\lim_{x \to \infty} f(x) = l$$

holds if and only if

$$\lim_{x \to +\infty} f(x) = \lim_{x \to -\infty} f(x) = l.$$

The usual uniqueness, arithmetic-operation, squeeze, and Cauchy properties remain valid in these forms whenever the corresponding expressions are meaningful.
A particularly important limit is

$$\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e.$$

4.6 Infinitesimals and Infinitely Large Functions

Definition 4.6.1. If

$$\lim_{x \to x_0} f(x) = 0,$$

then f is called an infinitesimal as $x \to x_0$.

Definition 4.6.2. If for every $A > 0$, there exists $\delta > 0$ such that

$$0 < |x - x_0| < \delta$$

implies

$$|f(x)| > A,$$

then f is called infinitely large as $x \to x_0$.

Infinitesimals can be compared by their quotients.
Suppose f and g are infinitesimals as $x \to x_0$.

Definition 4.6.3. If

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 0,$$

then f is called an infinitesimal of higher order than g, written as

$$f = o(g) \qquad (x \to x_0).$$

Definition 4.6.4. If

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = c, \qquad c \neq 0,$$

then f and g are called infinitesimals of the same order.

In particular, if

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = 1,$$

then f and g are called equivalent infinitesimals, written as

$$f \sim g \qquad (x \to x_0).$$

For example,

$$\sin x \sim x \qquad (x \to 0),$$

$$1 - \cos x \sim \frac{x^2}{2} \qquad (x \to 0).$$

Theorem 4.6.5. Equivalent infinitesimals may replace each other in products and quotients when computing limits.

The notation

$$f = O(g) \qquad (x \to x_0)$$

means that there exists a constant $M > 0$ such that

$$|f(x)| \leq M\,|g(x)|$$

for all x sufficiently close to x_0.
The notation

$$f = o(g)$$

means

$$\frac{f(x)}{g(x)} \to 0.$$

4.7 Continuous Functions

Definition 4.7.1. Let f be defined in a neighborhood of x_0. The function f is called continuous at x_0 if

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Equivalently, f is continuous at x_0 if for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$|x - x_0| < \delta$$

implies

$$|f(x) - f(x_0)| < \varepsilon.$$

Example 4.7.2. Every constant function is continuous.

Example 4.7.3. The function

$$f(x) = x$$

is continuous on $\mathbb{R}$.

The functions

$$\sin x \qquad \text{and} \qquad \cos x$$

are continuous on $\mathbb{R}$.

Definition 4.7.4. A function f is right-continuous at x_0 if

$$\lim_{x \to x_0+} f(x) = f(x_0).$$

It is left-continuous at x_0 if

$$\lim_{x \to x_0-} f(x) = f(x_0).$$

A function is continuous at x_0 if and only if it is both left-continuous and right-continuous at x_0.

Theorem 4.7.5. Suppose f and g are continuous at x_0. Then

$$f + g, \qquad fg$$

are continuous at x_0. If $g(x_0) \neq 0$, then

$$\frac{f}{g}$$

is also continuous at x_0.

Theorem 4.7.6. Suppose g is continuous at x_0, and f is continuous at $g(x_0)$. Then the composite function

$$f \circ g$$

is continuous at x_0.

Proof. Let

$$y_0 = g(x_0).$$

Since f is continuous at y_0, for every $\varepsilon > 0$, there exists $\eta > 0$ such that

$$|y - y_0| < \eta \Longrightarrow |f(y) - f(y_0)| < \varepsilon.$$

Since g is continuous at x_0, there exists $\delta > 0$ such that

$$|x - x_0| < \delta \Longrightarrow |g(x) - g(x_0)| < \eta.$$

Therefore

$$|f(g(x)) - f(g(x_0))| < \varepsilon,$$

so $f \circ g$ is continuous at x_0. □

Theorem 4.7.7. Let f be strictly monotone and continuous on an interval I. Then its inverse function f^{-1} is continuous on $f(I)$.

4.8 Discontinuities

If f is not continuous at x_0, then x_0 is called a discontinuity of f.

Definition 4.8.1. Suppose f has finite one-sided limits at x_0, but f is not continuous at x_0. Then x_0 is called a discontinuity of the first kind.

There are two cases.
If

$$\lim_{x\to x_0-} f(x) = \lim_{x\to x_0+} f(x),$$

but this common value is different from $f(x_0)$, or $f(x_0)$ is not defined, then x_0 is a removable discontinuity.
If

$$\lim_{x\to x_0-} f(x) \neq \lim_{x\to x_0+} f(x),$$

then x_0 is a jump discontinuity.

Definition 4.8.2. If at least one of the two one-sided limits at x_0 does not exist or is infinite, then x_0 is called a discontinuity of the second kind.

Theorem 4.8.3. A monotone function on an interval can have only discontinuities of the first kind.

4.9 Continuous Functions and Limit Computation

If f is continuous at x_0, then

$$\lim_{x\to x_0} f(x) = f(x_0).$$

Thus, when a function is continuous, taking a limit and applying the function may be interchanged:

$$\lim_{x\to x_0} f(g(x)) = f\left(\lim_{x\to x_0} g(x)\right),$$

provided the corresponding continuity assumptions are satisfied.
One important limit is

$$\lim_{x\to 0} \frac{\ln(1+x)}{x} = 1.$$

From this limit, one obtains

$$\lim_{x\to 0} \frac{a^x - 1}{x} = \ln a, \qquad a > 0.$$

Also,

$$\lim_{x\to 0} \frac{(1+x)^\alpha - 1}{x} = \alpha.$$

These limits, together with equivalent infinitesimals and suitable changes of variables, provide standard methods for computing many limits.

4.10 Uniform Continuity

Let I be an interval.

Definition 4.10.1. A function $f : I \to \mathbb{R}$ is called uniformly continuous on I if for every $\varepsilon > 0$, there exists $\delta > 0$ such that for any $x_1, x_2 \in I$,

$$|x_1 - x_2| < \delta$$

implies

$$|f(x_1) - f(x_2)| < \varepsilon.$$

The essential difference between continuity and uniform continuity is that in the definition of uniform continuity, δ depends only on ε, and not on the particular point of the interval.

Example 4.10.2. The function

$$f(x) = \sqrt{x}$$

is uniformly continuous on $[0, +\infty)$.

Proof. For $x_1, x_2 \geq 0$,

$$|\sqrt{x_1} - \sqrt{x_2}| \leq \sqrt{|x_1 - x_2|}.$$

Given $\varepsilon > 0$, choose

$$\delta = \varepsilon^2.$$

Then

$$|x_1 - x_2| < \delta$$

implies

$$|\sqrt{x_1} - \sqrt{x_2}| < \varepsilon.$$

□

The function

$$f(x) = x^2$$

is not uniformly continuous on $[0, +\infty)$.
The function

$$f(x) = \sin \frac{1}{x}$$

is not uniformly continuous on $(0, 1)$.

Example 4.10.3. The function

$$f(x) = \frac{1}{x}$$

is uniformly continuous on $[a, +\infty)$ for every $a > 0$.

4.11 Continuous Functions on a Closed Interval

We now consider a function

$$f : [a, b] \to \mathbb{R}$$

that is continuous on the closed interval $[a, b]$.

Theorem 4.11.1. Every continuous function on a closed interval is uniformly continuous.

Proof. Suppose f is not uniformly continuous on $[a, b]$. Then there exists $\varepsilon_0 > 0$ such that for every $n \in \mathbb{N}^+$, there are $x_n, y_n \in [a, b]$ satisfying

$$|x_n - y_n| < \frac{1}{n}$$

but

$$|f(x_n) - f(y_n)| \geq \varepsilon_0.$$

The sequence $\{x_n\}$ is bounded. By the Bolzano–Weierstrass theorem, it has a convergent subsequence

$$x_{n_k} \to x^* \in [a, b].$$

Since

$$|x_{n_k} - y_{n_k}| \to 0,$$

we also have

$$y_{n_k} \to x^*.$$

By continuity,

$$f(x_{n_k}) \to f(x^*)$$

and

$$f(y_{n_k}) \to f(x^*),$$

which implies

$$|f(x_{n_k}) - f(y_{n_k})| \to 0.$$

This contradicts

$$|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0.$$

Hence f is uniformly continuous. □

Theorem 4.11.2 (Boundedness Theorem)**.** Every continuous function on a closed interval is bounded.

Proof. Suppose f is unbounded on $[a, b]$. Then for every $n \in \mathbb{N}^+$, there exists $x_n \in [a, b]$ such that

$$|f(x_n)| > n.$$

By the Bolzano–Weierstrass theorem, $\{x_n\}$ has a convergent subsequence

$$x_{n_k} \to x^* \in [a, b].$$

Continuity gives

$$f(x_{n_k}) \to f(x^*),$$

so $\{f(x_{n_k})\}$ must be bounded, a contradiction. □

Theorem 4.11.3 (Extreme Value Theorem)**.** If f is continuous on $[a,b]$, then there exist $x^*, x_* \in [a,b]$ such that

$$f(x^*) = \max_{x\in[a,b]} f(x)$$

and

$$f(x_*) = \min_{x\in[a,b]} f(x).$$

Proof. Let

$$M = \sup_{x\in[a,b]} f(x).$$

By the definition of supremum, there exists a sequence $\{x_n\} \subset [a,b]$ such that

$$M - \frac{1}{n} < f(x_n) \le M.$$

By Bolzano–Weierstrass, a subsequence satisfies

$$x_{n_k} \to x^* \in [a,b].$$

By continuity,

$$f(x^*) = \lim_{k\to\infty} f(x_{n_k}) = M.$$

Thus the maximum is attained. The minimum is proved similarly. □

Theorem 4.11.4 (Intermediate Value Theorem)**.** Suppose f is continuous on $[a,b]$, and

$$f(a)f(b) < 0.$$

Then there exists $c \in (a,b)$ such that

$$f(c) = 0.$$

Proof. Suppose

$$f(a) < 0 < f(b).$$

Bisect $[a,b]$. Choose the half interval whose endpoint values have opposite signs. Repeating this procedure gives nested closed intervals

$$[a_1,b_1] \supset [a_2,b_2] \supset \cdots$$

such that

$$f(a_n) \le 0, \qquad f(b_n) \ge 0,$$

and

$$b_n - a_n = \frac{b-a}{2^{n-1}} \to 0.$$

By the nested interval theorem, there exists c common to all these intervals, and

$$a_n \to c, \qquad b_n \to c.$$

By continuity,

$$f(a_n) \to f(c), \qquad f(b_n) \to f(c).$$

Since

$$f(a_n) \le 0 \le f(b_n),$$

we obtain

$$f(c) = 0.$$

□

Theorem 4.11.5 (Intermediate Value Property)**.** Suppose f is continuous on $[a, b]$. If γ lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that

$$f(c) = \gamma.$$

4.12 Upper and Lower Limits of Functions

Let f be defined in a deleted neighborhood of x_0. For $\delta > 0$, set

$$E_\delta = \{f(x) : 0 < |x - x_0| < \delta\}.$$

The upper and lower limits of f at x_0 are defined by

$$\limsup_{x \to x_0} f(x) = \lim_{\delta \to 0+} \sup E_\delta$$

and

$$\liminf_{x \to x_0} f(x) = \lim_{\delta \to 0+} \inf E_\delta.$$

Theorem 4.12.1. Let

$$a^* = \limsup_{x \to x_0} f(x).$$

Then:

1. there exists a sequence $x_n \to x_0$, $x_n \neq x_0$, such that

$$f(x_n) \to a^*;$$

2. for every $a > a^*$, there exists $\delta > 0$ such that

$$0 < |x - x_0| < \delta \Longrightarrow f(x) < a.$$

The corresponding statement holds for

$$\liminf_{x \to x_0} f(x).$$

Theorem 4.12.2. The finite limit

$$\lim_{x \to x_0} f(x) = a$$

exists if and only if

$$\limsup_{x \to x_0} f(x) = \liminf_{x \to x_0} f(x) = a.$$

If

$$f(x) \leq g(x)$$

near x_0, then

$$\liminf_{x \to x_0} f(x) \leq \liminf_{x \to x_0} g(x)$$

and

$$\limsup_{x \to x_0} f(x) \leq \limsup_{x \to x_0} g(x).$$

4.13 Iteration and Periodic Points

Let

$$f : I \to I$$

be a mapping from an interval into itself.
Define

$$f^2 = f \circ f, \qquad f^3 = f \circ f \circ f,$$

and, in general,

$$f^n = \underbrace{f \circ f \circ \cdots \circ f}_{n \text{ times}}.$$

Definition 4.13.1. A point $x \in I$ is called a periodic point of f if there is a positive integer m such that

$$f^m(x) = x.$$

If m is the smallest positive integer satisfying this equality, then m is called the period of x.
A point of period 1 is a fixed point:

$$f(x) = x.$$

If x has period m, then

$$x, f(x), f^2(x), \ldots, f^{m-1}(x)$$

form an m-cycle.

Theorem 4.13.2. Let $f : I \to I$ be continuous. If f has a periodic point of period 3, then f has periodic points of every positive integer period.

Sketch of Proof. Let x_0 be a point of period 3. Its orbit consists of three distinct points. Write them in increasing order as

$$\alpha < \beta < \gamma.$$

By considering the possible order of their images and repeatedly using the intermediate value theorem on suitable subintervals, one constructs points whose iterates move successively through prescribed subintervals and eventually return to the starting interval.
This construction yields a periodic point of every prescribed positive integer period. □

The relation between different periods can be described more precisely by the following ordering:

$$3 \prec 5 \prec 7 \prec 9 \prec \cdots$$

$$\prec 2 \cdot 3 \prec 2 \cdot 5 \prec 2 \cdot 7 \prec \cdots$$

$$\prec 2^2 \cdot 3 \prec 2^2 \cdot 5 \prec \cdots$$

$$\prec \cdots \prec 2^n \prec \cdots \prec 2^3 \prec 2^2 \prec 2 \prec 1.$$

Theorem 4.13.3 (Sharkovsky Theorem)**.** Let $f : I \to I$ be continuous. If f has a periodic point of period m, then it has a periodic point of period n for every n lying after m in the Sharkovsky ordering.

In particular, since 3 is the first number in this ordering, the existence of a period-3 point implies the existence of periodic points of every positive integer period.

Chapter 5

Derivatives

The derivative describes the local rate of change of a function. Its origin may be seen in problems concerning instantaneous velocity, linear density, and the slope of the tangent line to a curve. Although these problems have different interpretations, they lead to the same limiting process.

5.1 Definition of the Derivative

Let f be defined in a neighborhood of x_0.

Definition 5.1.1. If the finite limit

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

exists, then f is called differentiable at x_0, and this limit is called the derivative of f at x_0, denoted by

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}.$$

Equivalently, writing $h = x - x_0$,

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

Geometrically, if

$$y = f(x),$$

then $f'(x_0)$ is the slope of the tangent line at

$$P_0 = (x_0, f(x_0)).$$

Thus the tangent line is

$$y - f(x_0) = f'(x_0)(x - x_0).$$

Definition 5.1.2. The right derivative of f at x_0 is

$$f'_+(x_0) = \lim_{h \to 0+} \frac{f(x_0 + h) - f(x_0)}{h},$$

provided the limit exists.

Similarly, the left derivative is

$$f'_-(x_0) = \lim_{h \to 0-} \frac{f(x_0 + h) - f(x_0)}{h}.$$

The function f is differentiable at x_0 if and only if both one-sided derivatives exist and

$$f'_-(x_0) = f'_+(x_0).$$

Theorem 5.1.3. If f is differentiable at x_0, then f is continuous at x_0.

Proof. Since $f'(x_0)$ exists,

$$\lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0} = f'(x_0).$$

Hence

$$\lim_{x\to x_0}\big(f(x)-f(x_0)\big)=\lim_{x\to x_0}\frac{f(x)-f(x_0)}{x-x_0}(x-x_0)$$
$$=f'(x_0)\cdot 0=0.$$

Therefore

$$\lim_{x\to x_0}f(x)=f(x_0),$$

so f is continuous at x_0. □

The converse is not true. A continuous function need not be differentiable.
For example,

$$f(x)=|x|$$

is continuous at $x=0$, but

$$f'_-(0)=-1,\qquad f'_+(0)=1,$$

so $f'(0)$ does not exist.

Definition 5.1.4. If f is differentiable at every point of an interval I, then f is called differentiable on I.

5.2 Basic Derivatives

For a constant c,

$$(c)'=0.$$

For $n\in\mathbb{N}^+$,

$$(x^n)'=nx^{n-1}.$$

More generally,

$$(x^\alpha)'=\alpha x^{\alpha-1}$$

where the expression is defined.
For the trigonometric functions,

$$(\sin x)'=\cos x,$$
$$(\cos x)'=-\sin x,$$
$$(\tan x)'=\frac{1}{\cos^2 x},$$

and

$$(\cot x)'=-\frac{1}{\sin^2 x}.$$

For exponential functions,

$$(a^x)'=a^x\ln a,\qquad a>0.$$

In particular,

$$(e^x)'=e^x.$$

For logarithmic functions,

$$(\log_a x)'=\frac{1}{x\ln a},\qquad a>0,\quad a\neq 1,$$

and hence

$$(\ln x)'=\frac{1}{x}.$$

The derivatives of the inverse trigonometric functions include

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, \qquad |x| < 1,$$

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}},$$

and

$$(\arctan x)' = \frac{1}{1+x^2}.$$

5.3 Rules of Differentiation

Suppose f and g are differentiable at x.
Then

$$(f+g)'(x) = f'(x) + g'(x),$$

and

$$(f-g)'(x) = f'(x) - g'(x).$$

The product rule is

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

If $g(x) \neq 0$, then the quotient rule is

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}.$$

Proof of the Product Rule. We have

$$\begin{aligned}&\frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &\quad = \frac{f(x+h)-f(x)}{h}g(x+h) + f(x)\frac{g(x+h)-g(x)}{h}.\end{aligned}$$

Since differentiability implies continuity,

$$g(x+h) \to g(x).$$

Letting $h \to 0$, we obtain

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

□

Theorem 5.3.1 (Chain Rule)**.** Suppose $y = f(u)$ is differentiable at u_0, and $u = \varphi(x)$ is differentiable at x_0, where

$$u_0 = \varphi(x_0).$$

Then the composite function

$$y = f(\varphi(x))$$

is differentiable at x_0, and

$$(f \circ \varphi)'(x_0) = f'(\varphi(x_0))\varphi'(x_0).$$

Thus, using differential notation for the derivative,

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

For several successive compositions,

$$y = f(u), \qquad u = \varphi(v), \qquad v = \psi(x),$$

we obtain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dv}\frac{dv}{dx}.$$

5.4 Derivative of an Inverse Function

Theorem 5.4.1. Suppose $y = f(x)$ is strictly monotone and continuous in a neighborhood of x_0, and f is differentiable at x_0 with

$$f'(x_0) \neq 0.$$

If the inverse function $x = f^{-1}(y)$ is defined near

$$y_0 = f(x_0),$$

then f^{-1} is differentiable at y_0, and

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

Proof. Since

$$y = f(x), \qquad y_0 = f(x_0),$$

we have

$$\frac{x - x_0}{y - y_0} = \left(\frac{f(x) - f(x_0)}{x - x_0}\right)^{-1}.$$

Letting $y \to y_0$, equivalently $x \to x_0$, gives

$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)}.$$

□

For example, since

$$y = \sin x$$

and

$$\frac{dy}{dx} = \cos x,$$

we obtain

$$(\arcsin y)' = \frac{1}{\sqrt{1 - y^2}}.$$

5.5 Higher-Order Derivatives

If f' is differentiable, its derivative is called the second derivative of f, written as

$$f''(x)$$

or

$$f^{(2)}(x).$$

Inductively, the n-th derivative is denoted by

$$f^{(n)}(x).$$

For example,

$$\frac{d^n}{dx^n}e^x = e^x.$$

For trigonometric functions, the derivatives repeat periodically:

$$\sin x, \quad \cos x, \quad -\sin x, \quad -\cos x, \quad \sin x, \ldots.$$

Theorem 5.5.1 (Leibniz Formula)**.** Suppose f and g possess derivatives up to order n. Then

$$(fg)^{(n)} = \sum_{k=0}^{n} \binom{n}{k} f^{(k)} g^{(n-k)}.$$

Thus

$$\begin{aligned}(fg)' &= f'g + fg',\\ (fg)'' &= f''g + 2f'g' + fg'',\\ (fg)''' &= f'''g + 3f''g' + 3f'g'' + fg'''.\end{aligned}$$

5.6 Local Extrema and Fermat's Theorem

Definition 5.6.1. Suppose f is defined in a neighborhood of x_0. If there exists $\delta > 0$ such that

$$f(x) \le f(x_0)$$

whenever

$$|x - x_0| < \delta,$$

then $f(x_0)$ is called a local maximum of f.

Similarly, if

$$f(x) \ge f(x_0)$$

for all x sufficiently close to x_0, then $f(x_0)$ is called a local minimum.
Local maxima and local minima are collectively called local extrema.

Theorem 5.6.2 (Fermat's Theorem)**.** Suppose f has a local extremum at an interior point x_0, and $f'(x_0)$ exists. Then

$$f'(x_0) = 0.$$

Proof. Suppose $f(x_0)$ is a local maximum. For sufficiently small $h > 0$,

$$f(x_0 + h) - f(x_0) \le 0,$$

so

$$\frac{f(x_0+h)-f(x_0)}{h} \le 0.$$

Hence

$$f'_+(x_0) \le 0.$$

For sufficiently small $h < 0$,

$$f(x_0+h)-f(x_0) \le 0,$$

and therefore

$$\frac{f(x_0+h)-f(x_0)}{h} \ge 0.$$

Thus

$$f'_-(x_0) \ge 0.$$

Since the derivative exists,

$$f'_-(x_0) = f'_+(x_0),$$

and hence

$$f'(x_0) = 0.$$

The proof for a local minimum is similar. □

A point satisfying

$$f'(x_0) = 0$$

is called a stationary point.

5.7 Mean Value Theorems

Theorem 5.7.1 (Rolle's Theorem)**.** Suppose f is continuous on $[a,b]$, differentiable on (a,b), and

$$f(a) = f(b).$$

Then there exists $\xi \in (a,b)$ such that

$$f'(\xi) = 0.$$

Proof. Since f is continuous on $[a,b]$, it attains its maximum and minimum.
If both are equal, f is constant and

$$f'(x) = 0$$

for every $x \in (a,b)$.
Otherwise, at least one of the maximum and minimum is attained at an interior point $\xi \in (a,b)$. By Fermat's theorem,

$$f'(\xi) = 0.$$

□

Theorem 5.7.2 (Lagrange Mean Value Theorem)**.** Suppose f is continuous on $[a,b]$ and differentiable on (a,b). Then there exists $\xi \in (a,b)$ such that

$$f'(\xi) = \frac{f(b)-f(a)}{b-a}.$$

Equivalently,

$$f(b)-f(a) = f'(\xi)(b-a).$$

Proof. Define

$$F(x) = f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then

$$F(a) = F(b).$$

By Rolle's theorem, there exists $\xi \in (a, b)$ such that

$$F'(\xi) = 0.$$

Therefore

$$f'(\xi) - \frac{f(b) - f(a)}{b - a} = 0,$$

which proves the result. □

A useful form is

$$f(x) - f(x_0) = f'(\xi)(x - x_0),$$

where ξ lies between x and x_0.

Corollary 5.7.3. If

$$f'(x) = 0$$

for every $x \in (a, b)$, then f is constant on (a, b).

Corollary 5.7.4. If

$$|f'(x)| \leq M$$

on an interval, then

$$|f(x) - f(y)| \leq M\,|x - y|\,.$$

Theorem 5.7.5 (Cauchy Mean Value Theorem)**.** Suppose f and g are continuous on $[a, b]$ and differentiable on (a, b). Then there exists $\xi \in (a, b)$ such that

$$\Big(f(b) - f(a)\Big)g'(\xi) = \Big(g(b) - g(a)\Big)f'(\xi).$$

If, in addition,

$$g'(\xi) \neq 0$$

and

$$g(b) \neq g(a),$$

then

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)}.$$

Proof. Define

$$\begin{aligned} F(x) = &\Big(f(b) - f(a)\Big)\Big(g(x) - g(a)\Big) \\ &- \Big(g(b) - g(a)\Big)\Big(f(x) - f(a)\Big). \end{aligned}$$

Then

$$F(a) = F(b) = 0.$$

By Rolle's theorem, there exists $\xi \in (a, b)$ such that

$$F'(\xi) = 0.$$

Hence

$$\Big(f(b) - f(a)\Big)g'(\xi) = \Big(g(b) - g(a)\Big)f'(\xi).$$

□

5.8 Darboux Theorem for Derivatives

Although a derivative need not be continuous, it has the intermediate value property.

Theorem 5.8.1 (Darboux Theorem)**.** Suppose f is differentiable on $[a, b]$ and

$$f'(a) < \lambda < f'(b).$$

Then there exists $\xi \in (a, b)$ such that

$$f'(\xi) = \lambda.$$

Proof. Define

$$F(x) = f(x) - \lambda x.$$

Then

$$F'(a) = f'(a) - \lambda < 0$$

and

$$F'(b) = f'(b) - \lambda > 0.$$

Since F is continuous on $[a, b]$, it attains a minimum at some point $\xi \in [a, b]$.
The inequalities

$$F'(a) < 0, \qquad F'(b) > 0$$

show that the minimum cannot occur at either endpoint. Hence

$$\xi \in (a, b).$$

By Fermat's theorem,

$$F'(\xi) = 0.$$

Therefore

$$f'(\xi) = \lambda.$$

□

Thus derivatives have the intermediate value property even when they are not continuous.

5.9 Monotonicity

Theorem 5.9.1. Suppose f is continuous on $[a, b]$ and differentiable on (a, b).
If

$$f'(x) \geq 0$$

for every $x \in (a, b)$, then f is increasing on $[a, b]$.
If

$$f'(x) \leq 0,$$

then f is decreasing on $[a, b]$.

Proof. Let

$$a \leq x_1 < x_2 \leq b.$$

By the Lagrange mean value theorem, there exists $\xi \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(\xi)(x_2 - x_1).$$

If $f'(\xi) \geq 0$, then

$$f(x_2) \geq f(x_1).$$

Hence f is increasing. □

If

$$f'(x) > 0$$

throughout (a, b), then f is strictly increasing.
Similarly, if

$$f'(x) < 0,$$

then f is strictly decreasing.
The converse for strict monotonicity is not generally true: a strictly increasing function may have derivative zero at some points.

5.10 Local Extrema from the Sign of the Derivative

Suppose f is continuous near x_0 and differentiable on both sides of x_0.
If

$$f'(x) > 0$$

to the left of x_0, and

$$f'(x) < 0$$

to the right of x_0, then $f(x_0)$ is a local maximum.
If

$$f'(x) < 0$$

to the left of x_0, and

$$f'(x) > 0$$

to the right of x_0, then $f(x_0)$ is a local minimum.
Thus the sign change of the first derivative determines local extrema.

5.11 Convexity and Concavity

Definition 5.11.1. A function f is called convex on an interval I if for any $x_1, x_2 \in I$ and $0 \leq \lambda \leq 1$,

$$f\Big(\lambda x_1 + (1-\lambda)x_2\Big) \leq \lambda f(x_1) + (1-\lambda) f(x_2).$$

If the inequality is reversed, f is called concave.
Geometrically, for a convex function, the graph lies below every chord joining two points of the graph.

Theorem 5.11.2. Suppose f is differentiable on an interval I. Then f is convex on I if and only if f' is increasing on I.

Theorem 5.11.3. Suppose f is twice differentiable on an interval I. If

$$f''(x) \geq 0$$

for every $x \in I$, then f is convex on I.
If

$$f''(x) \leq 0,$$

then f is concave on I.

Definition 5.11.4. A point at which the graph changes from convex to concave or from concave to convex is called an inflection point.

5.12 Jensen Inequality

Theorem 5.12.1 (Jensen Inequality)**.** Suppose f is convex on an interval I. Let

$$x_1, \ldots, x_n \in I$$

and let

$$\lambda_1, \ldots, \lambda_n \geq 0, \qquad \lambda_1 + \cdots + \lambda_n = 1.$$

Then

$$f\left(\sum_{k=1}^{n} \lambda_k x_k\right) \leq \sum_{k=1}^{n} \lambda_k f(x_k).$$

In the special case

$$\lambda_1 = \cdots = \lambda_n = \frac{1}{n},$$

we obtain

$$f\left(\frac{x_1 + \cdots + x_n}{n}\right) \leq \frac{f(x_1) + \cdots + f(x_n)}{n}.$$

For a concave function, the inequalities are reversed.

5.13 L'Hospital's Rule

L'Hospital's rule is used to calculate certain limits of indeterminate forms.

Theorem 5.13.1 (L'Hospital's Rule, 0/0 Form)**.** Suppose f and g are differentiable on a deleted neighborhood of x_0, and

$$\lim_{x \to x_0} f(x) = \lim_{x \to x_0} g(x) = 0.$$

Assume

$$g'(x) \neq 0$$

near x_0. If

$$\lim_{x \to x_0} \frac{f'(x)}{g'(x)} = A,$$

where A may be finite or infinite, then

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = A.$$

Sketch of Proof. For x sufficiently close to x_0, apply the Cauchy mean value theorem to f and g on the interval joining x and x_0. There exists ξ between x and x_0 such that

$$\frac{f(x) - f(x_0)}{g(x) - g(x_0)} = \frac{f'(\xi)}{g'(\xi)}.$$

Using the limiting assumptions and letting $x \to x_0$, we obtain the desired result. □

Theorem 5.13.2 (L'Hospital's Rule, ∞/∞ Form)**.** Suppose

$$|f(x)| \to \infty, \qquad |g(x)| \to \infty,$$

as $x \to x_0$, and the remaining assumptions of L'Hospital's rule are satisfied. If

$$\lim_{x \to x_0} \frac{f'(x)}{g'(x)} = A,$$

then

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = A.$$

The same method applies to limits as

$$x \to +\infty \qquad \text{or} \qquad x \to -\infty.$$

L'Hospital's rule applies directly only to the forms

$$\frac{0}{0} \qquad \text{and} \qquad \frac{\infty}{\infty}.$$

Other indeterminate forms should first be transformed into one of these two forms. Such forms include

$$0 \cdot \infty, \qquad \infty - \infty, \qquad 1^{\infty}, \qquad 0^0, \qquad \infty^0.$$

For an exponential form

$$y = f(x)^{g(x)},$$

it is often convenient to take logarithms:

$$\ln y = g(x) \ln f(x).$$

5.14 Taylor Formula

Let f possess sufficiently many derivatives near x_0. We seek a polynomial

$$P_n(x) = a_0 + a_1(x - x_0) + \cdots + a_n(x - x_0)^n$$

such that

$$P_n^{(k)}(x_0) = f^{(k)}(x_0), \qquad k = 0, 1, \ldots, n.$$

These conditions determine

$$a_k = \frac{f^{(k)}(x_0)}{k!}.$$

Hence the corresponding polynomial is

$$\begin{aligned} P_n(x) = f(x_0) &+ f'(x_0)(x - x_0) \\ &+ \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n. \end{aligned}$$

Theorem 5.14.1 (Taylor Formula with Peano Remainder)**.** Suppose f has derivatives up to order n at x_0. Then

$$\begin{aligned} f(x) = f(x_0) &+ f'(x_0)(x - x_0) \\ &+ \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots \\ &+ \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + o\Big((x - x_0)^n\Big) \end{aligned}$$

as $x \to x_0$.

When $x_0 = 0$, the formula is called the Maclaurin formula:

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + o(x^n).$$

The standard Maclaurin formulas include

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + o(x^n),$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots,$$

and

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots,$$

with the corresponding remainder terms.
Also,

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1}\frac{x^n}{n} + o(x^n),$$

as $x \to 0$.

5.15 Taylor Formula with Lagrange Remainder

Theorem 5.15.1 (Taylor Formula with Lagrange Remainder)**.** Suppose f has derivatives up to order $n+1$ on an interval containing x_0 and x. Then

$$\begin{aligned} f(x) = f(x_0) &+ f'(x_0)(x - x_0) \\ &+ \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots \\ &+ \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n + R_n(x), \end{aligned}$$

where

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$$

for some ξ between x_0 and x.

The term

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$$

is called the Lagrange remainder.
For the Maclaurin formula,

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}x^{n+1},$$

where ξ lies between 0 and x.

5.16 Taylor Formula with Cauchy Remainder

Under the corresponding assumptions, the Taylor remainder can also be written in the Cauchy form

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{n!}(x - \xi)^n(x - x_0),$$

where ξ lies between x_0 and x.
Thus the three remainder forms used in these notes are:

$$R_n(x) = o\Big((x - x_0)^n\Big)$$

for the Peano remainder,

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x - x_0)^{n+1}$$

for the Lagrange remainder, and

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{n!}(x - \xi)^n(x - x_0)$$

for the Cauchy remainder.
Taylor's formula provides both a local polynomial approximation to a function and a systematic method for evaluating limits and estimating errors.

Chapter 6

Differentials

The derivative describes the rate of change of a function, while the differential describes the principal linear part of its increment. For a function of one variable, differentiability and the existence of a derivative are equivalent, but the differential provides another useful way to express the local behavior of a function.

6.1 Differentiability and Differentials

Let

$$y = f(x),$$

and let x have an increment Δx. Then the corresponding increment of y is

$$\Delta y = f(x + \Delta x) - f(x).$$

Definition 6.1.1. The function f is said to be differentiable at x if its increment can be written in the form

$$\Delta y = A\Delta x + o(\Delta x), \qquad \Delta x \to 0,$$

where A is independent of Δx.

The term

$$A\Delta x$$

is the principal linear part of the increment Δy.

Theorem 6.1.2. A function f is differentiable at x if and only if $f'(x)$ exists. In this case,

$$A = f'(x).$$

Proof. Suppose first that f is differentiable at x. Then

$$\Delta y = A\Delta x + o(\Delta x).$$

For $\Delta x \neq 0$,

$$\frac{\Delta y}{\Delta x} = A + \frac{o(\Delta x)}{\Delta x}.$$

Letting $\Delta x \to 0$, we obtain

$$f'(x) = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = A.$$

Conversely, suppose $f'(x)$ exists. Then

$$\frac{\Delta y}{\Delta x} = f'(x) + o(1), \qquad \Delta x \to 0.$$

Multiplying by Δx, we obtain

$$\Delta y = f'(x)\Delta x + o(\Delta x).$$

Hence f is differentiable at x. □

Definition 6.1.3. If f is differentiable at x, then the quantity

$$f'(x)\Delta x$$

is called the differential of $y = f(x)$, denoted by dy. Thus

$$dy = f'(x)\Delta x.$$

For the independent variable x,

$$dx = \Delta x.$$

Therefore

$$\boxed{dy = f'(x)\,dx}$$

and consequently

$$f'(x) = \frac{dy}{dx}.$$

Thus the increment of a differentiable function satisfies

$$\Delta y = dy + o(\Delta x).$$

Equivalently,

$$\Delta y = f'(x)\Delta x + o(\Delta x).$$

6.2 Geometric Meaning of the Differential

Consider the curve

$$y = f(x)$$

and a point

$$P = (x, f(x)).$$

When the independent variable changes from x to $x + \Delta x$, the actual change in the ordinate of the curve is

$$\Delta y = f(x + \Delta x) - f(x).$$

The tangent line to the curve at P has slope $f'(x)$. Therefore the corresponding change in the ordinate along the tangent line is

$$f'(x)\Delta x.$$

Hence

$$dy = f'(x)\Delta x$$

is geometrically the increment in the ordinate of the tangent line corresponding to the horizontal increment Δx.
Since

$$\Delta y = dy + o(\Delta x),$$

the differential dy gives the principal linear approximation to the actual increment Δy.

6.3 Differential Rules

Let

$$u = u(x), \qquad v = v(x)$$

be differentiable functions.
For a constant c,

$$d(c) = 0.$$

For the independent variable,

$$d(x) = dx.$$

The differential of a sum is

$$d(u + v) = du + dv.$$

Similarly,

$$d(u - v) = du - dv.$$

For a constant c,

$$d(cu) = c\,du.$$

The product rule is

$$d(uv) = u\,dv + v\,du.$$

If $v \neq 0$, then

$$d\left(\frac{u}{v}\right) = \frac{v\,du - u\,dv}{v^2}.$$

For a composite function

$$y = f(u), \qquad u = \varphi(x),$$

we have

$$dy = f'(u)\,du$$

and

$$du = \varphi'(x)\,dx.$$

Therefore

$$dy = f'(u)\varphi'(x)\,dx.$$

This agrees with the chain rule

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

6.4 Implicit Differentiation

Some relations between x and y are given implicitly rather than in the explicit form $y = f(x)$.
Suppose x and y satisfy an equation

$$F(x, y) = 0.$$

If y may be regarded as a differentiable function of x, we differentiate both sides with respect to x, regarding y as a function of x.
For example, let

$$x^2 + y^2 = 1.$$

Taking differentials gives

$$2x\,dx + 2y\,dy = 0.$$

Hence

$$2y\,dy = -2x\,dx,$$

and therefore

$$\frac{dy}{dx} = -\frac{x}{y}, \qquad y \neq 0.$$

Thus an explicit formula for y is not always necessary in order to calculate its derivative.

6.5 Parametric Differentiation

Suppose a curve is given parametrically by

$$x = \varphi(t), \qquad y = \psi(t).$$

Then

$$dx = \varphi'(t)\,dt$$

and

$$dy = \psi'(t)\,dt.$$

Therefore, whenever

$$\varphi'(t) \neq 0,$$

we have

$$\boxed{\frac{dy}{dx} = \frac{\psi'(t)}{\varphi'(t)}}$$

For example, consider the ellipse

$$x = a\cos t, \qquad y = b\sin t.$$

Then

$$\frac{dx}{dt} = -a\sin t$$

and

$$\frac{dy}{dt} = b\cos t.$$

Hence

$$\frac{dy}{dx} = \frac{b\cos t}{-a\sin t} = -\frac{b}{a}\cot t.$$

The same rule allows derivatives to be calculated directly from parametric equations without first eliminating the parameter.

Chapter 7

Foundations of Integration

Differentiation assigns a derivative to a given function. Integration begins with the inverse problem: given a function f, find a function whose derivative is f.
This point of view leads first to antiderivatives and indefinite integrals. Definite integrals arise from a different problem: approximating a quantity by finite sums and passing to a limit. The fundamental theorem of calculus establishes the connection between these two notions.

7.1 Antiderivatives and Indefinite Integrals

Definition 7.1.1. Let f be defined on an interval I. If a differentiable function F satisfies

$$F'(x) = f(x), \qquad x \in I,$$

then F is called an antiderivative of f on I.

If F is an antiderivative of f, then for every constant C,

$$(F(x) + C)' = f(x).$$

Thus $F(x) + C$ is also an antiderivative of f.
Conversely, any two antiderivatives of the same function on an interval differ by a constant.

Theorem 7.1.2. Suppose F and G are antiderivatives of f on an interval I. Then there exists a constant C such that

$$F(x) - G(x) = C, \qquad x \in I.$$

Proof. Since

$$F'(x) = G'(x) = f(x),$$

we have

$$(F - G)'(x) = 0.$$

By the Lagrange mean value theorem, a function whose derivative vanishes on an interval is constant. Hence

$$F(x) - G(x) = C.$$

□

Definition 7.1.3. The collection of all antiderivatives of f is called the indefinite integral of f, denoted by

$$\int f(x)\,dx.$$

If $F'(x) = f(x)$, then

$$\int f(x)\,dx = F(x) + C.$$

Here C is called the constant of integration.
It is important to distinguish an indefinite integral from a definite integral. The expression

$$\int f(x)\,dx$$

denotes a family of antiderivatives, whereas

$$\int_a^b f(x)\,dx$$

will be defined later as a number obtained from a limiting process.

7.2 Basic Integration Formulas

The basic formulas for indefinite integration follow directly from the corresponding derivative formulas.

For a constant c,

$$\int c\,dx = cx + C.$$

For $\alpha \neq -1$,

$$\int x^\alpha\,dx = \frac{x^{\alpha+1}}{\alpha+1} + C.$$

In particular,

$$\int \frac{1}{x}\,dx = \ln|x| + C.$$

For exponential functions,

$$\int e^x\,dx = e^x + C,$$

and

$$\int a^x\,dx = \frac{a^x}{\ln a} + C, \qquad a > 0, \quad a \neq 1.$$

For trigonometric functions,

$$\int \sin x\,dx = -\cos x + C,$$

$$\int \cos x\,dx = \sin x + C,$$

$$\int \frac{1}{\cos^2 x}\,dx = \tan x + C,$$

and

$$\int \frac{1}{\sin^2 x}\,dx = -\cot x + C.$$

Also,

$$\int \frac{1}{1+x^2}\,dx = \arctan x + C,$$

and

$$\int \frac{1}{\sqrt{1-x^2}}\,dx = \arcsin x + C.$$

The linearity of indefinite integration gives

$$\int \Big(af(x) + bg(x)\Big)\,dx = a\int f(x)\,dx + b\int g(x)\,dx.$$

7.3 Integration by Parts

The product rule

$$(uv)' = u'v + uv'$$

gives

$$u\,dv = d(uv) - v\,du.$$

Integrating both sides yields the formula for integration by parts.

Theorem 7.3.1 (Integration by Parts)**.** If the expressions involved are defined, then

$$\boxed{\int u\,dv = uv - \int v\,du}$$

or equivalently,

$$\boxed{\int u(x)v'(x)\,dx = u(x)v(x) - \int u'(x)v(x)\,dx.}$$

For example,

$$\begin{aligned}\int xe^x\,dx &= xe^x - \int e^x\,dx \\ &= xe^x - e^x + C.\end{aligned}$$

Repeated integration by parts may be used for integrals such as

$$\int x^n e^x\,dx.$$

Another standard example is

$$\begin{aligned}\int \ln x\,dx &= x\ln x - \int x\frac{1}{x}\,dx \\ &= x\ln x - x + C.\end{aligned}$$

7.4 Change of Variables

Suppose

$$u = \varphi(x).$$

Then formally,

$$du = \varphi'(x)\,dx.$$

This gives the basic substitution formula.

Theorem 7.4.1 (Change of Variables)**.** If $F'(u) = f(u)$, then

$$\int f(\varphi(x))\varphi'(x)\,dx = F(\varphi(x)) + C.$$

Equivalently,

$$\boxed{\int f(\varphi(x))\varphi'(x)\,dx = \int f(u)\,du.}$$

For example,

$$\int 2x\cos(x^2)\,dx.$$

Let

$$u = x^2.$$

Then

$$du = 2x\,dx,$$

and therefore

$$\int 2x\cos(x^2)\,dx = \int \cos u\,du = \sin u + C = \sin(x^2) + C.$$

Substitutions involving trigonometric functions are also useful. For example, expressions containing

$$\sqrt{a^2 - x^2}$$

may be transformed by setting

$$x = a\sin t.$$

7.5 Integration of Rational Functions

A rational function has the form

$$R(x) = \frac{P(x)}{Q(x)},$$

where P and Q are polynomials.
If

$$\deg P \geq \deg Q,$$

polynomial division is first used to write

$$R(x) = S(x) + \frac{P_1(x)}{Q(x)},$$

where

$$\deg P_1 < \deg Q.$$

The proper rational function

$$\frac{P_1(x)}{Q(x)}$$

can then be decomposed into partial fractions according to the factorization of $Q(x)$.
For example,

$$\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}.$$

After determining A and B, the integral is reduced to basic forms involving logarithmic and inverse trigonometric functions.

7.6 The Definite Integral

We now introduce a different notion of integration.
Let f be a bounded function on $[a, b]$. Divide the interval by points

$$a = x_0 < x_1 < \cdots < x_n = b.$$

Write

$$\Delta x_i = x_i - x_{i-1}, \qquad i = 1, \ldots, n,$$

and choose

$$\xi_i \in [x_{i-1}, x_i].$$

The corresponding Riemann sum is

$$\sum_{i=1}^{n} f(\xi_i)\Delta x_i.$$

Let

$$\lambda = \max_{1 \leq i \leq n} \Delta x_i.$$

Definition 7.6.1. If there exists a real number I such that

$$\lim_{\lambda \to 0} \sum_{i=1}^{n} f(\xi_i)\Delta x_i = I,$$

independently of the choice of the partition and of the points ξ_i, then f is called Riemann integrable on $[a, b]$. The number I is called the definite integral of f over $[a, b]$, written as

$$\int_a^b f(x)\, dx = I.$$

Thus the definite integral is defined by a limit of finite sums, rather than as a family of antiderivatives.
By convention,

$$\int_a^a f(x)\,dx = 0$$

and

$$\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx.$$

7.7 Basic Properties of Definite Integrals

Suppose f and g are integrable on $[a, b]$.
For constants α, β,

$$\int_a^b \Big(\alpha f(x) + \beta g(x)\Big)\,dx = \alpha \int_a^b f(x)\,dx + \beta \int_a^b g(x)\,dx.$$

If $c \in [a, b]$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx.$$

If

$$f(x) \le g(x)$$

on $[a, b]$, then

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx.$$

In particular, if

$$f(x) \ge 0,$$

then

$$\int_a^b f(x)\,dx \ge 0.$$

Also,

$$\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx.$$

If

$$m \le f(x) \le M$$

on $[a, b]$, then

$$m(b-a) \le \int_a^b f(x)\,dx \le M(b-a).$$

7.8 Symmetry in Definite Integrals

Symmetry often simplifies the computation of definite integrals.
If f is odd, then

$$f(-x) = -f(x),$$

and

$$\boxed{\int_{-a}^a f(x)\,dx = 0.}$$

If f is even, then

$$f(-x) = f(x),$$

and

$$\boxed{\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx.}$$

More generally, under the substitution

$$x = a + b - t,$$

we obtain

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx.$$

Hence

$$\int_a^b f(x)\,dx = \frac{1}{2}\int_a^b \Big(f(x) + f(a+b-x)\Big)\,dx.$$

7.9 The Fundamental Theorem of Calculus

The definite integral and the indefinite integral are defined in different ways. Their fundamental relation is established by the fundamental theorem of calculus.

Theorem 7.9.1. Suppose f is continuous on $[a, b]$. Define

$$F(x) = \int_a^x f(t)\,dt.$$

Then F is differentiable on (a, b), and

$$F'(x) = f(x).$$

Proof. For sufficiently small h,

$$\begin{aligned} F(x+h) - F(x) &= \int_a^{x+h} f(t)\,dt - \int_a^x f(t)\,dt \\ &= \int_x^{x+h} f(t)\,dt. \end{aligned}$$

By the mean value theorem for integrals, there exists a point ξ_h between x and $x+h$ such that

$$\int_x^{x+h} f(t)\,dt = f(\xi_h)h.$$

Hence

$$\frac{F(x+h) - F(x)}{h} = f(\xi_h).$$

As $h \to 0$,

$$\xi_h \to x.$$

By continuity of f,

$$f(\xi_h) \to f(x).$$

Therefore

$$F'(x) = f(x).$$

□

Theorem 7.9.2 (Newton–Leibniz Formula)**.** Suppose f is continuous on $[a, b]$, and F is an antiderivative of f. Then

$$\boxed{\int_a^b f(x)\,dx = F(b) - F(a).}$$

Proof. Let

$$G(x) = \int_a^x f(t)\,dt.$$

By the fundamental theorem of calculus,

$$G'(x) = f(x) = F'(x).$$

Hence

$$G(x) - F(x)$$

is constant. Therefore

$$G(b) - G(a) = F(b) - F(a).$$

Since

$$G(a) = 0, \qquad G(b) = \int_a^b f(x)\,dx,$$

the result follows. □

7.10 Change of Variables and Integration by Parts

Theorem 7.10.1 (Change of Variables for Definite Integrals)**.** Under the corresponding continuity and differentiability conditions, if

$$x = \varphi(t),$$

then

$$\boxed{\int_a^b f(x)\,dx = \int_\alpha^\beta f(\varphi(t))\varphi'(t)\,dt,}$$

where

$$\varphi(\alpha) = a, \qquad \varphi(\beta) = b.$$

Unlike an indefinite integral, the limits of integration must also be changed when the variable is changed.

Theorem 7.10.2 (Integration by Parts for Definite Integrals)**.** If u and v are continuously differentiable on $[a, b]$, then

$$\boxed{\int_a^b u(x)v'(x)\,dx = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x)\,dx.}$$

Here

$$[F(x)]_a^b = F(b) - F(a).$$

7.11 Darboux Sums and Integrability

Let

$$P : a = x_0 < x_1 < \cdots < x_n = b$$

be a partition of $[a, b]$.
For each subinterval

$$[x_{i-1}, x_i],$$

let

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x)$$

and

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x).$$

Definition 7.11.1. The upper Darboux sum of f corresponding to P is

$$U(P, f) = \sum_{i=1}^{n} M_i \Delta x_i.$$

The lower Darboux sum is

$$L(P, f) = \sum_{i=1}^{n} m_i \Delta x_i.$$

Clearly,

$$L(P, f) \leq U(P, f).$$

If a partition is refined, the lower sum cannot decrease and the upper sum cannot increase.

Theorem 7.11.2 (Darboux Criterion)**.** A bounded function f on $[a, b]$ is Riemann integrable if and only if for every $\varepsilon > 0$, there exists a partition P such that

$$U(P, f) - L(P, f) < \varepsilon.$$

Since

$$U(P, f) - L(P, f) = \sum_{i=1}^{n} (M_i - m_i) \Delta x_i,$$

the criterion measures the total oscillation of f over the partition.

Theorem 7.11.3. Every continuous function on a closed interval is Riemann integrable.

Sketch of Proof. A continuous function on $[a, b]$ is uniformly continuous. Given $\varepsilon > 0$, choose the partition sufficiently fine so that the oscillation of f on each subinterval is sufficiently small. Then

$$U(P, f) - L(P, f) < \varepsilon.$$

The Darboux criterion therefore gives the integrability of f. □

Theorem 7.11.4. Every monotone function on a closed interval is Riemann integrable.

Sketch of Proof. Suppose f is increasing. For an equal partition,

$$U(P, f) - L(P, f) = \sum_{i=1}^{n} \Big(f(x_i) - f(x_{i-1}) \Big) \Delta x.$$

Since the sum of the differences telescopes,

$$U(P, f) - L(P, f) = \Big(f(b) - f(a) \Big) \Delta x.$$

This tends to zero as the mesh tends to zero. Hence f is integrable. □

7.12 Lebesgue Criterion for Riemann Integrability

The preceding results give sufficient conditions for integrability. The textbook also gives a characterization in terms of discontinuities.

Theorem 7.12.1 (Lebesgue Criterion)**.** A bounded function on $[a, b]$ is Riemann integrable if and only if its set of discontinuities has measure zero.

In this chapter, this result is used only as a criterion for Riemann integrability. No general theory of Lebesgue integration is developed here.
In particular, a bounded function with only finitely many discontinuities is Riemann integrable.
More generally, the criterion also applies when the set of discontinuities has measure zero.

7.13 Improper Integrals over Unbounded Intervals

The Riemann integral introduced above is defined for bounded functions on bounded intervals. The definition can be extended by a limiting process.

Definition 7.13.1. Suppose f is integrable on every finite interval $[a, A]$, where $A > a$. If the finite limit

$$\lim_{A\to+\infty} \int_a^A f(x)\,dx$$

exists, then the improper integral

$$\int_a^{+\infty} f(x)\,dx$$

is said to converge, and

$$\boxed{\int_a^{+\infty} f(x)\,dx = \lim_{A\to+\infty} \int_a^A f(x)\,dx.}$$

If the limit does not exist as a finite number, the improper integral is said to diverge.
Similarly,

$$\int_{-\infty}^b f(x)\,dx = \lim_{A\to-\infty} \int_A^b f(x)\,dx,$$

provided the limit exists.
For an integral over the whole real line, choose a finite number c and write

$$\int_{-\infty}^{+\infty} f(x)\,dx = \int_{-\infty}^c f(x)\,dx + \int_c^{+\infty} f(x)\,dx.$$

The integral converges only when both integrals on the right converge.

Example 7.13.2. For $p > 0$, consider

$$\int_1^{+\infty} \frac{1}{x^p}\,dx.$$

If $p \neq 1$,

$$\begin{aligned}\int_1^A \frac{1}{x^p}\,dx &= \left[\frac{x^{1-p}}{1-p}\right]_1^A \\ &= \frac{A^{1-p}-1}{1-p}.\end{aligned}$$

Hence

$$\int_1^{+\infty} \frac{1}{x^p}\,dx$$

converges if $p > 1$ and diverges if $0 < p \leq 1$.

7.14 Improper Integrals of Unbounded Functions

Improper integrals also arise when the interval is bounded but the integrand is unbounded. Suppose f is integrable on every interval $[a, b-\varepsilon]$, but is unbounded near b.

Definition 7.14.1. If the finite limit

$$\lim_{\varepsilon \to 0+} \int_a^{b-\varepsilon} f(x)\,dx$$

exists, then the improper integral

$$\int_a^b f(x)\,dx$$

is said to converge, and its value is defined by this limit.

Similarly, if f is unbounded near a,

$$\int_a^b f(x)\,dx = \lim_{\varepsilon \to 0+} \int_{a+\varepsilon}^b f(x)\,dx,$$

provided the limit exists.
If f is unbounded at an interior point $c \in (a, b)$, then

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

and both improper integrals on the right must converge.

Example 7.14.2. For $p > 0$, consider

$$\int_0^1 \frac{1}{x^p}\,dx.$$

For $p \neq 1$,

$$\int_\varepsilon^1 x^{-p}\,dx = \frac{1-\varepsilon^{1-p}}{1-p}.$$

Therefore

$$\int_0^1 \frac{1}{x^p}\,dx$$

converges if $0 < p < 1$ and diverges if $p \geq 1$.

7.15 Convergence of Improper Integrals

Theorem 7.15.1 (Cauchy Criterion)**.** The improper integral

$$\int_a^{+\infty} f(x)\,dx$$

converges if and only if for every $\varepsilon > 0$, there exists $A > a$ such that for any $v > u > A$,

$$\left| \int_u^v f(x)\,dx \right| < \varepsilon.$$

Theorem 7.15.2 (Comparison Test)**.** Suppose

$$0 \leq f(x) \leq g(x)$$

for all sufficiently large x.
If

$$\int_a^{+\infty} g(x)\,dx$$

converges, then

$$\int_a^{+\infty} f(x)\,dx$$

also converges.
If

$$\int_a^{+\infty} f(x)\,dx$$

diverges, then

$$\int_a^{+\infty} g(x)\,dx$$

also diverges.

A corresponding comparison principle holds for improper integrals with an unbounded integrand.

Definition 7.15.3. An improper integral

$$\int_a^{+\infty} f(x)\,dx$$

is called absolutely convergent if

$$\int_a^{+\infty} |f(x)|\,dx$$

converges.

Theorem 7.15.4. Every absolutely convergent improper integral is convergent.

Thus absolute convergence is a sufficient condition for convergence of an improper integral.

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He also makes extensive use of examples and counterexamples. These examples do not simply demonstrate how a theorem is applied; they help clarify why each hypothesis is needed, where a conclusion may fail, and how related concepts should be distinguished. This style of teaching gradually taught me to pay attention not only to mathematical results, but also to the logical structure behind them.
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